

Computer Networks

Lecture 3

Shmulik Kaplan

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Application Layer

Examining popular application-level protocols

Lecture 3
- http, https, and the WWW in general (cookies, proxy, caching..)
- socket programming (only if there is time …)

Lecture 4
- smtp, pop, imap (email)
- dns
- content distribution networks (CDN)











Today

- Review
- URL
- Web and HTTP
- RTT
- Questions
- HTTP definition
  - Request & Response
  - Conditionals
  - Authentication
  - Cookies
- Demonstration
- Proxy
- Security
- HTTP 2 & HTTP 3

Network applications: some jargon

- A process: a program that is running within a host.
- Interprocess communication: communication between processes running on the same host.
- Application-layer protocol: communication between processes running in different hosts.

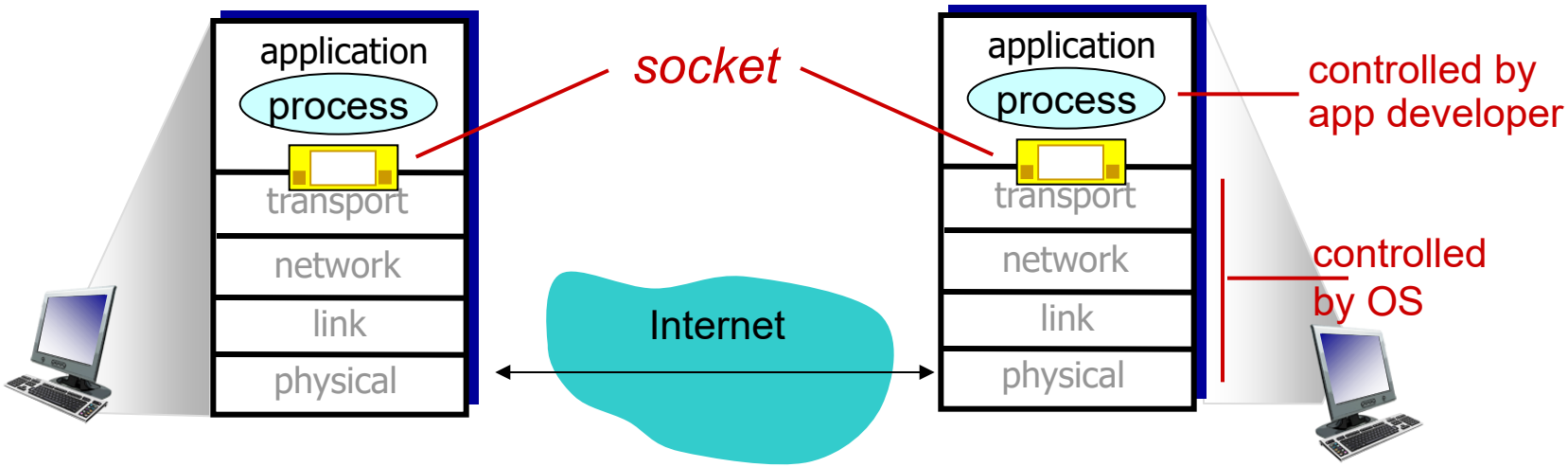


Socket-programming

Socket: a door between application process and end-end-transport protocol (UDP or TCP)

Port number: allows receiving host to determine to which local process the message should be delivered

Socket = IP address + port number



… lots more on this later

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Network apps



Application	Application layer protocol	Underlying transport protocol
e-mail	smtp [RFC 821]	TCP
remote terminal access	telnet [RFC 854]	TCP
Web	http [RFC 2068]	TCP
file transfer	ftp [RFC 959]	TCP
streaming multimedia	proprietary	typically, UDP
Internet telephony	proprietary	typically, UDP

URL – Uniform Resource Locator

- A form of addressing – enable defining a location of any resource on the web.
- Example: https://www.example.com:8080/path/to/resource?query=example#section
  - https:// ↔ protocol to use
  - www.example.com ↔ Server IP address
  - :8080 ↔ application port

  Socket
  - /path/to/resource ↔ path from root defined in server
  - ?query=example ↔ parameters for request
- Web server application defines its 'root' folder in its storage – from which the resources path is considered.

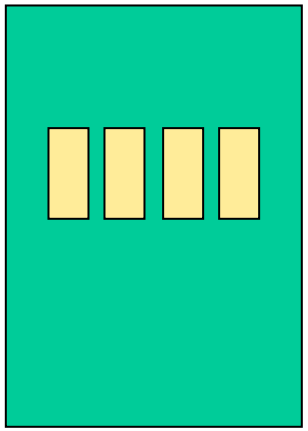
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Web and HTTP

First, a review…

- *web page* consists of *objects*
- object can be HTML file, JPEG image, audio file, video file, …
- web page consists of *base HTML-file* which includes *several referenced objects*
- each object is addressable by a *URL,* e.g.,



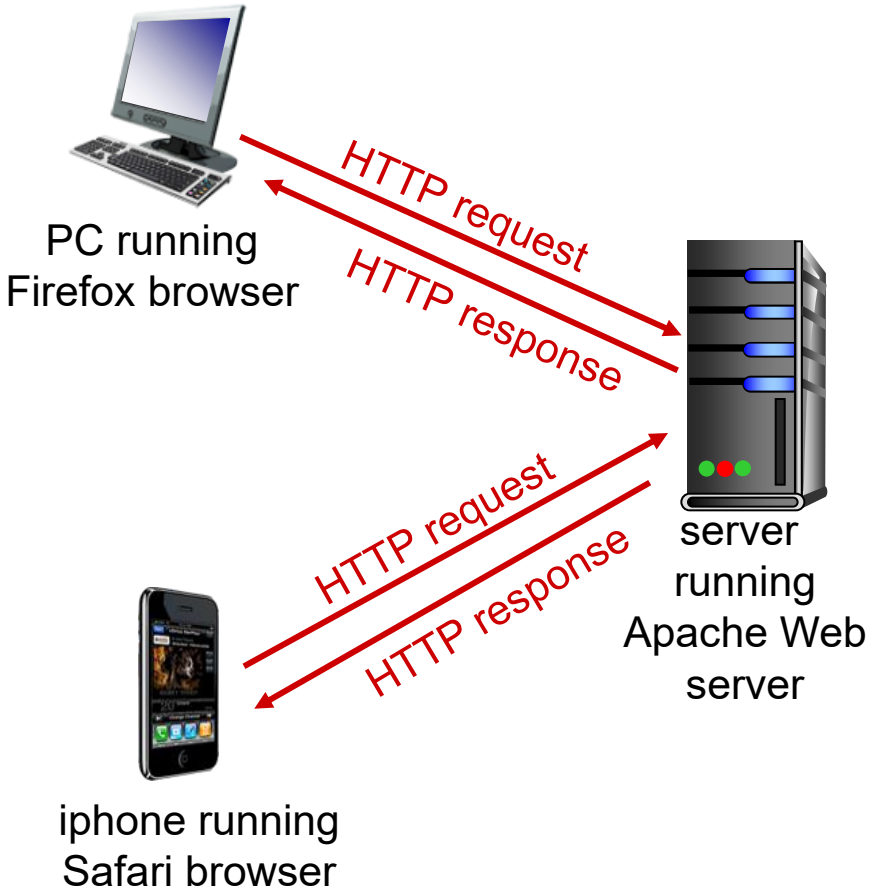
`www.someschool.edu/someDept/pic.gif`

host name | path name

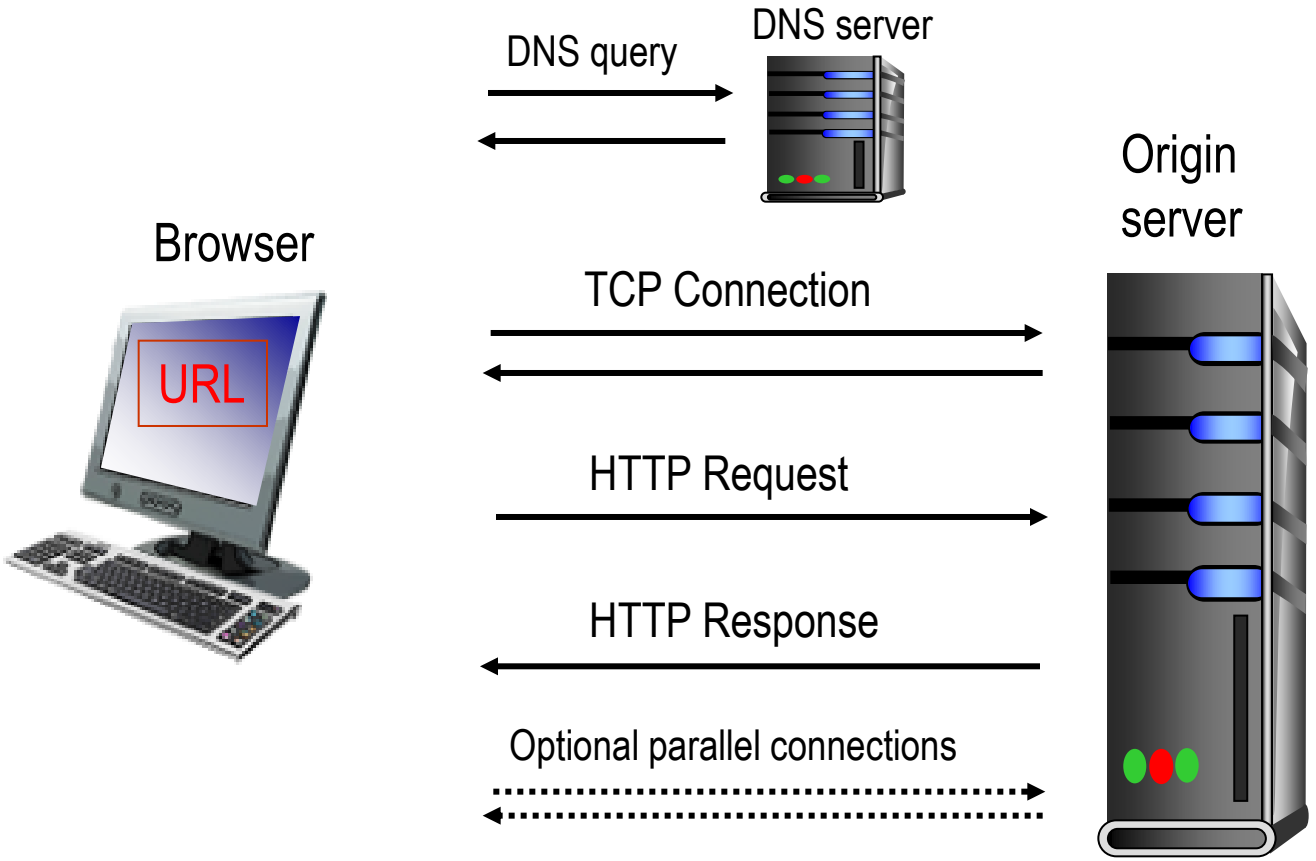
The Web: the http protocol

http: hypertext transfer protocol

- Web's application layer protocol
- client/server model
  - *client:* browser that requests, receives, "displays" Web objects
  - *server:* Web server sends objects in response to requests
- http/1.0: RFC 1945 (1996)
- http/1.1: RFC 2068 (1999)
- http/2 : RFC 7540 (2015)
- http/3 : RFC 9114 (June 2022)



Steps in a browser process



The http protocol: more

http: TCP transport service:

- client initiates TCP connection to server, port **80**
- server accepts TCP connection from client
- http messages (application-layer protocol messages) exchanged between browser (http client) and Web server (http server)
- TCP connection closed

http is "stateless"

- server maintains no information about past client requests

aside

Protocols that maintain "state" are complex!

- past history (state) must be maintained
- if server/client crashes, their views of "state" may be inconsistent, must be reconciled

HTTP connections

Non-persistent HTTP

- At most one object is sent over a TCP connection
- HTTP/1.0 uses nonpersistent HTTP

Persistent HTTP

- Multiple objects can be sent over single TCP connection between client and server
- HTTP/1.1 uses persistent connections in default mode

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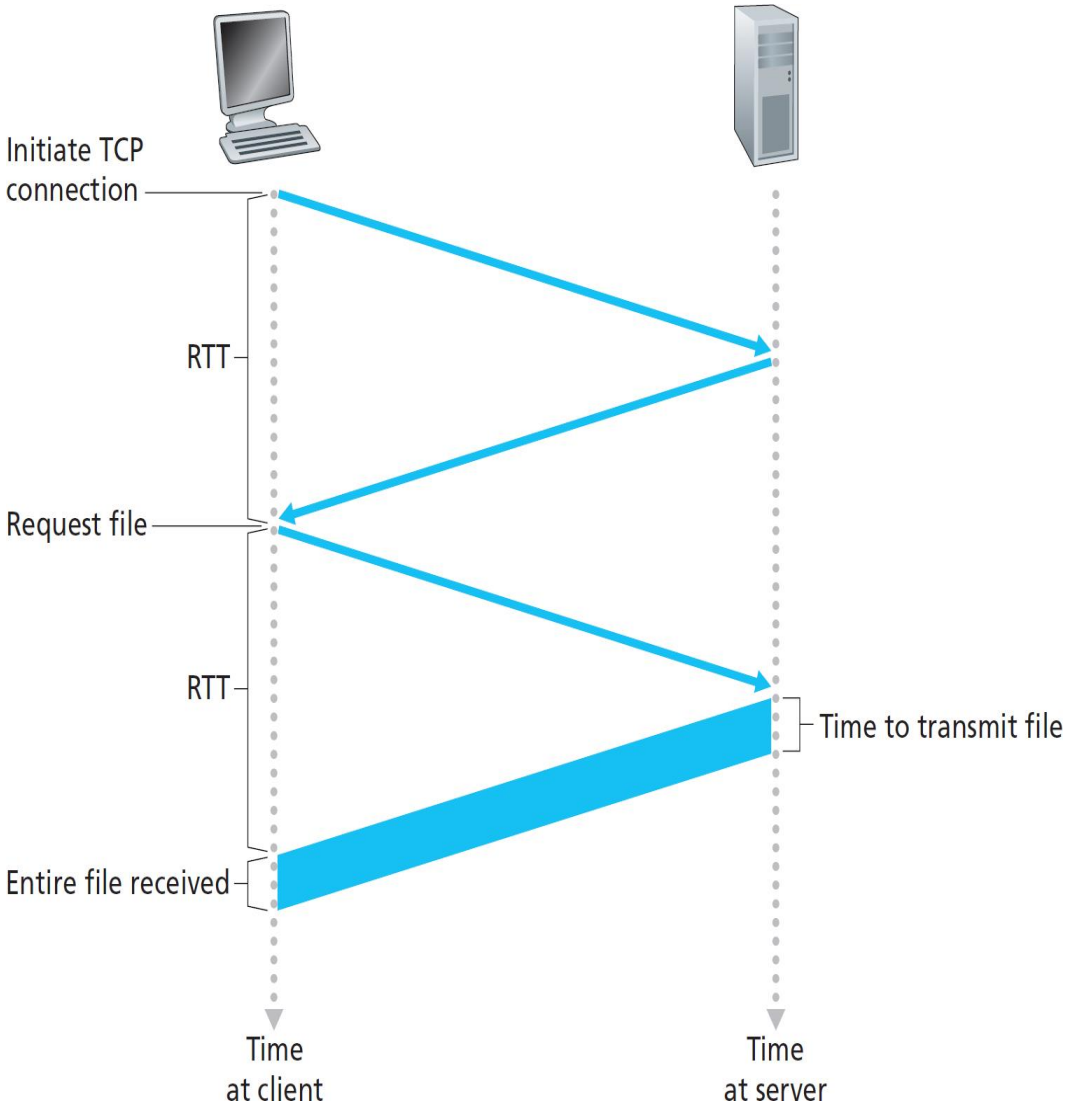
Response time modeling

Definition of RTT: time to send a small packet to travel from client to server and back.

Response time:

- one RTT to initiate TCP connection
- one RTT for HTTP request and first few bytes of HTTP response to return
- file transmission time

total = 2RTT+ transmit time



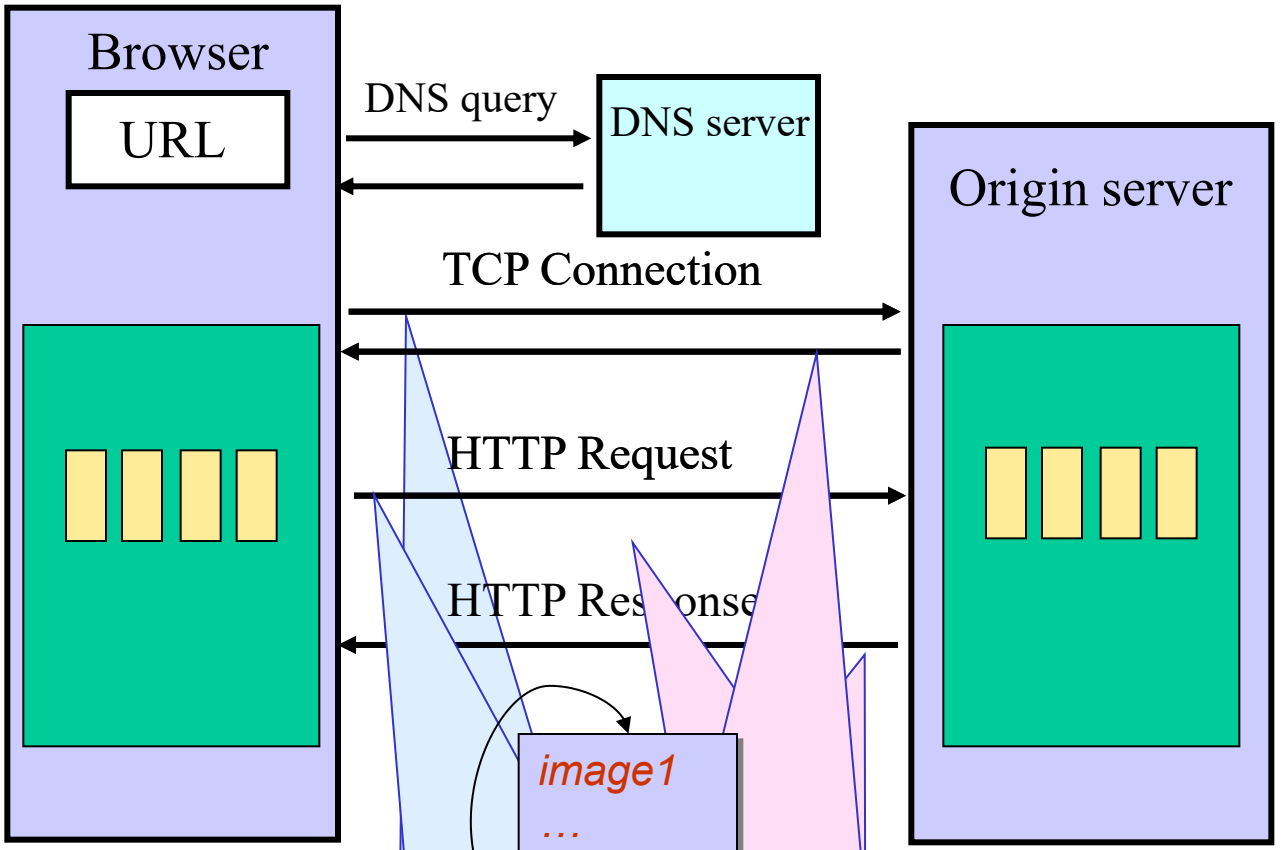
http 1.0 example

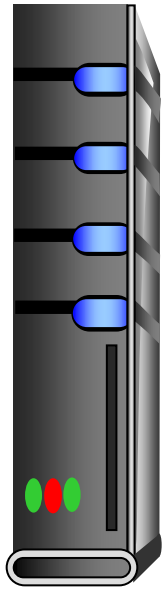
Non-persistent without parallel connections

Suppose user enters URL www.runi.ac.il/cs/index.html

(contains text, references to 4 images)





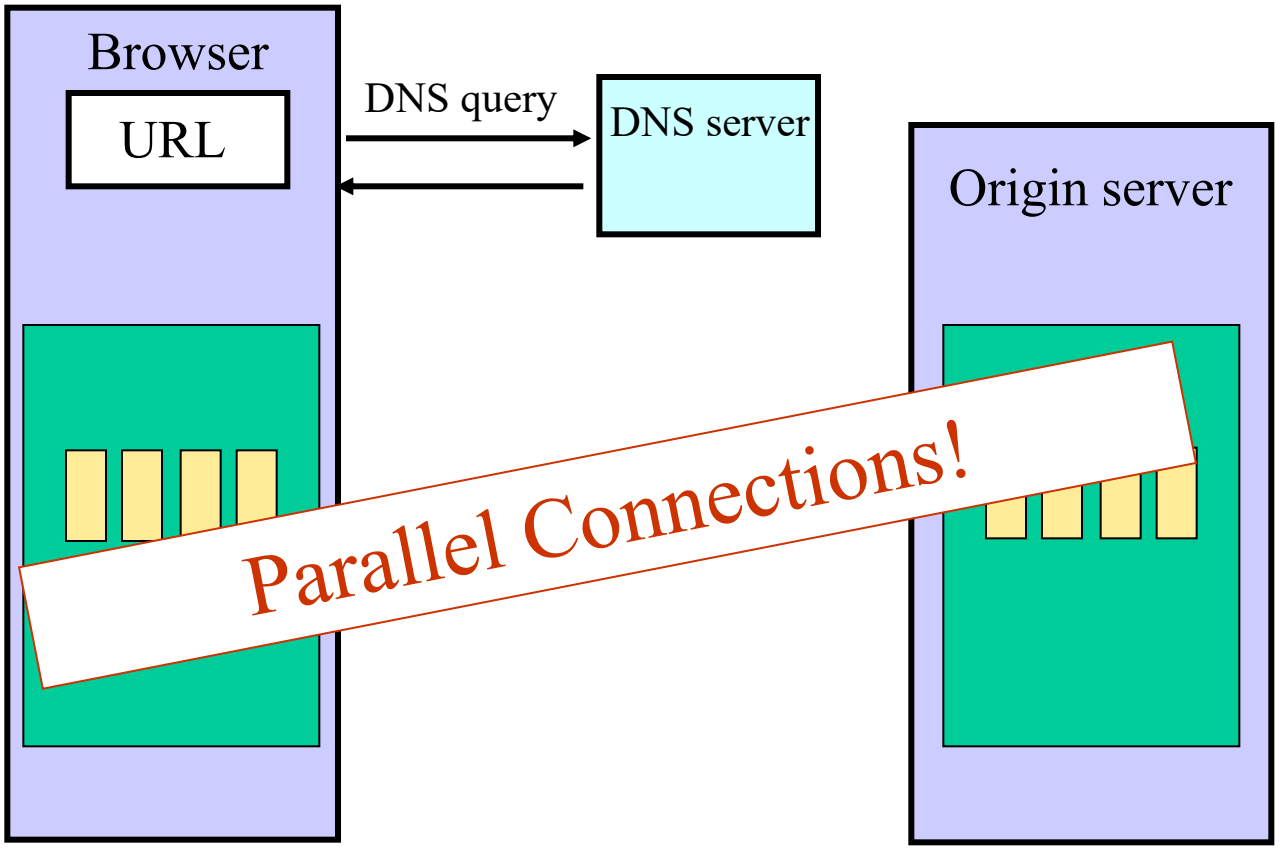


http 1.0 example

Non-persistent without parallel connections

Suppose user enters URL
www.idc.ac.il/cs/index.html

(contains text, references to 4 images)



Persistent HTTP

Non-persistent HTTP issues:

- requires 2 RTTs per object
- OS must work and allocate host resources for each TCP connection
- HTTP/1.0, but browsers often open parallel TCP connections to fetch referenced objects

Persistent HTTP

- server leaves connection open after sending response
- subsequent HTTP messages between same client/server are sent over connection

Persistent without pipelining:

- client issues new request only when previous response has been received
- one RTT for each referenced object

Persistent with pipelining:

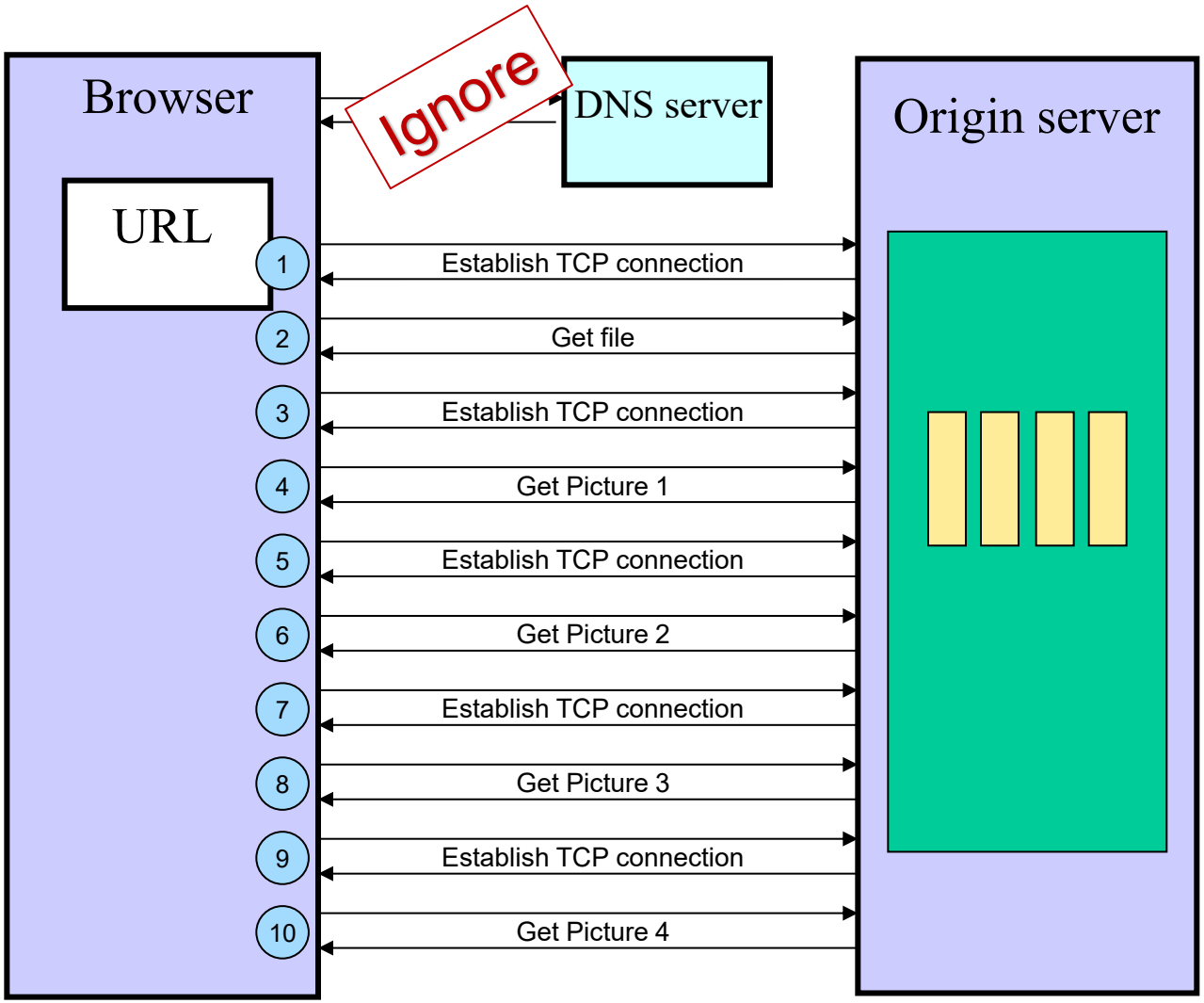
- client sends requests as soon as it encounters a referenced object
- as little as one RTT for all the referenced object
- Implementation problems … **HTTP/2**

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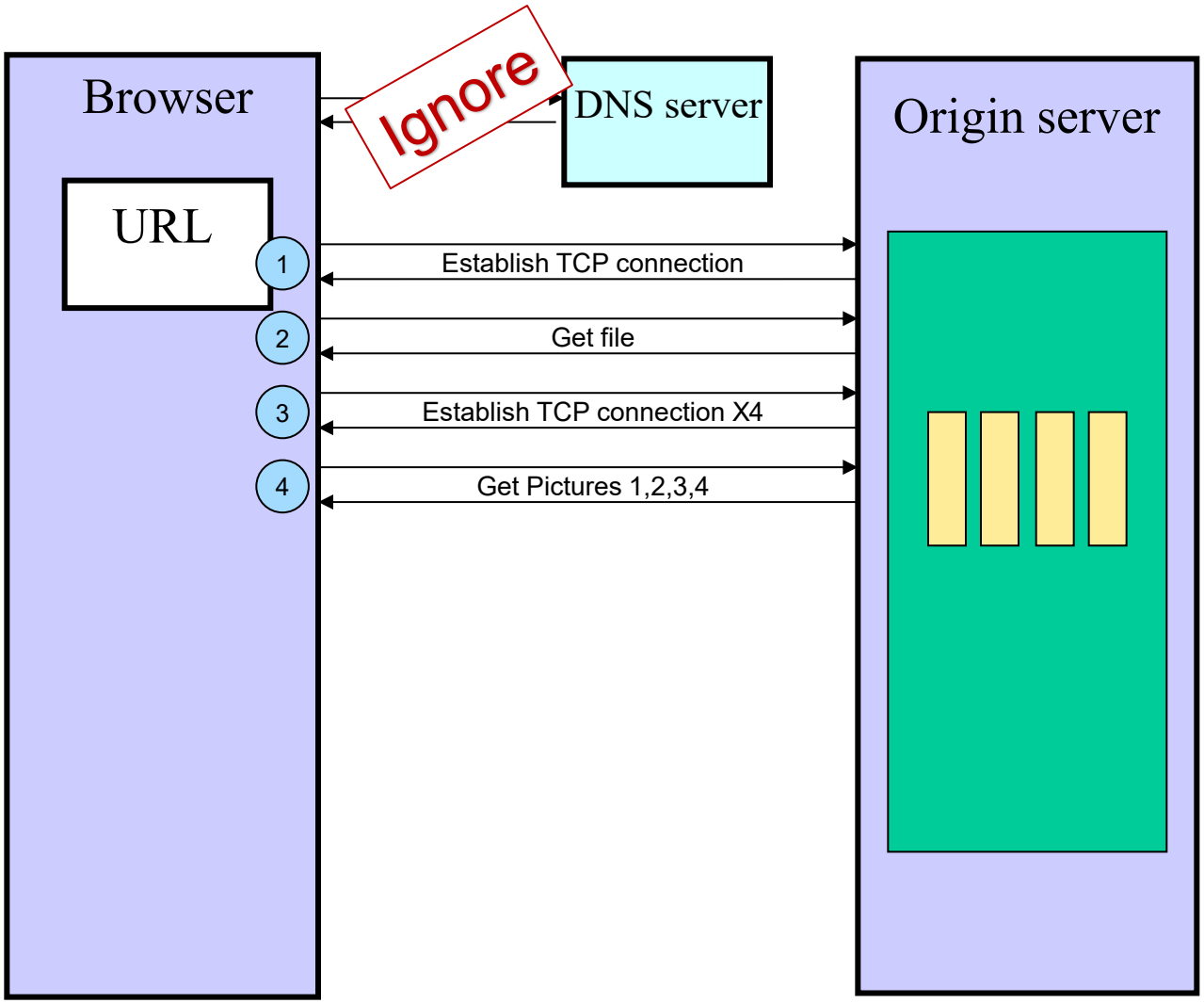
Questions

1. Non-persistent without parallel connections



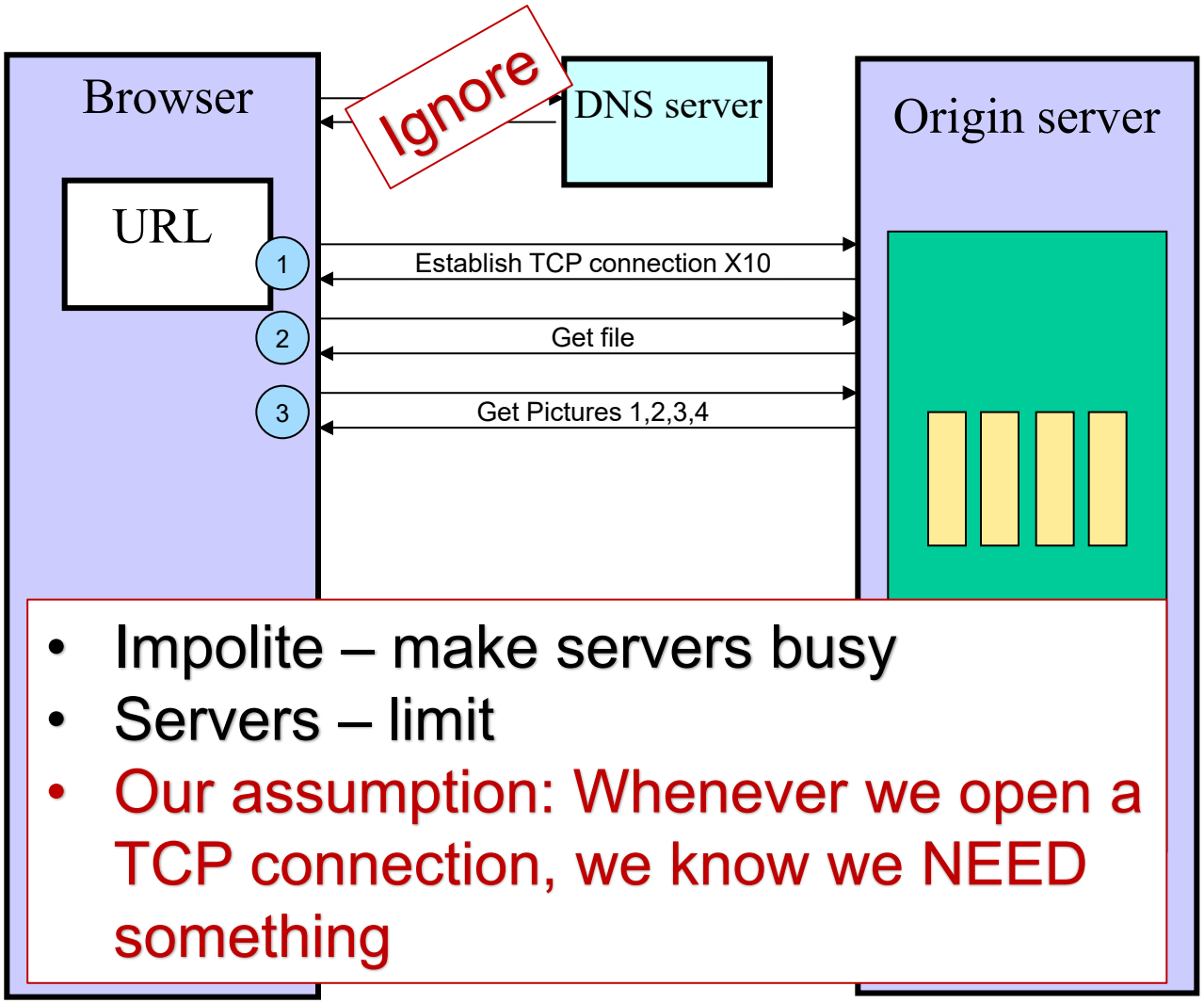
Questions

2. Non-persistent with parallel connections



Questions

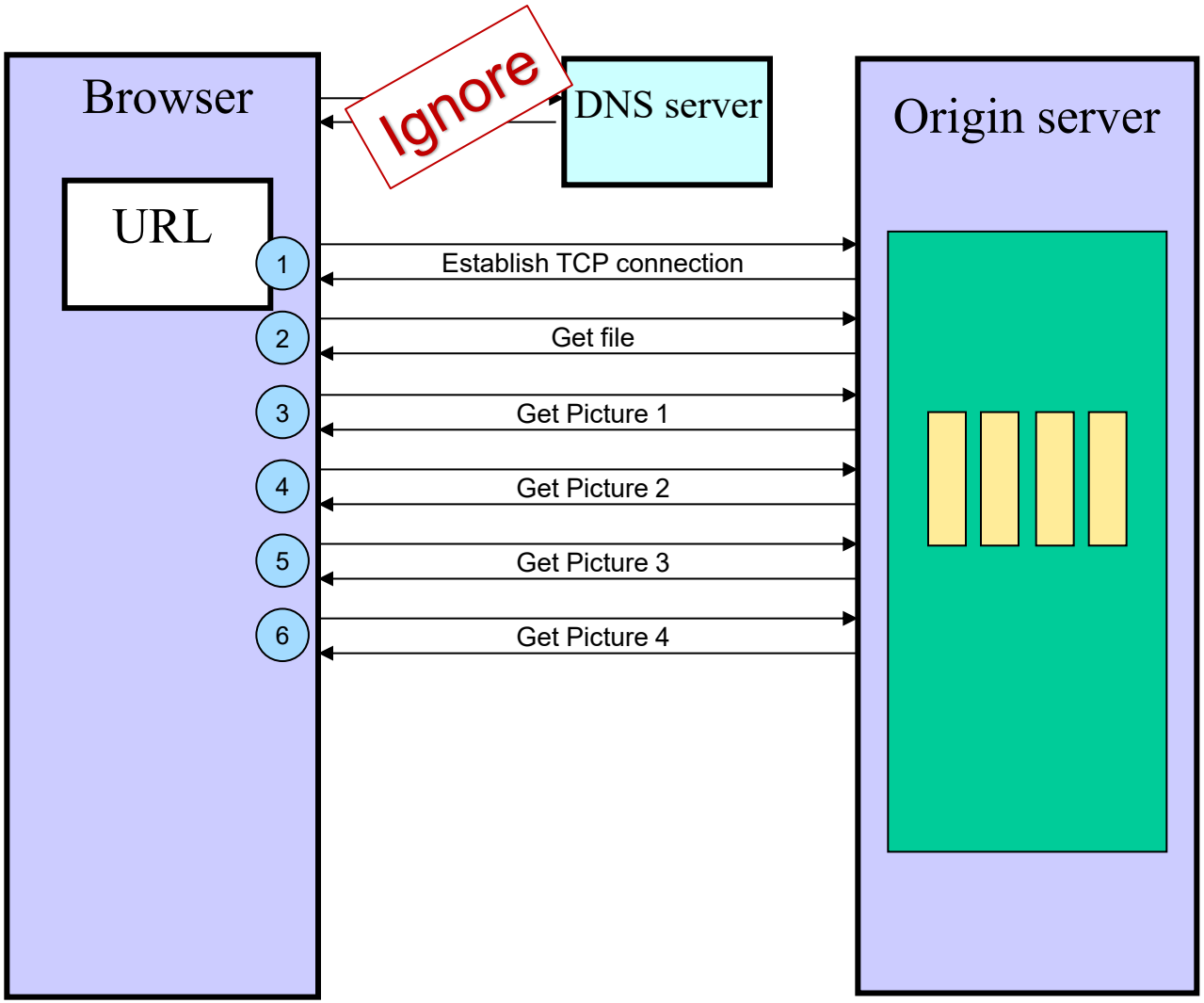
2. Non-persistent with parallel connections



- Impolite – make servers busy
- Servers – limit
- Our assumption: Whenever we open a TCP connection, we know we NEED something

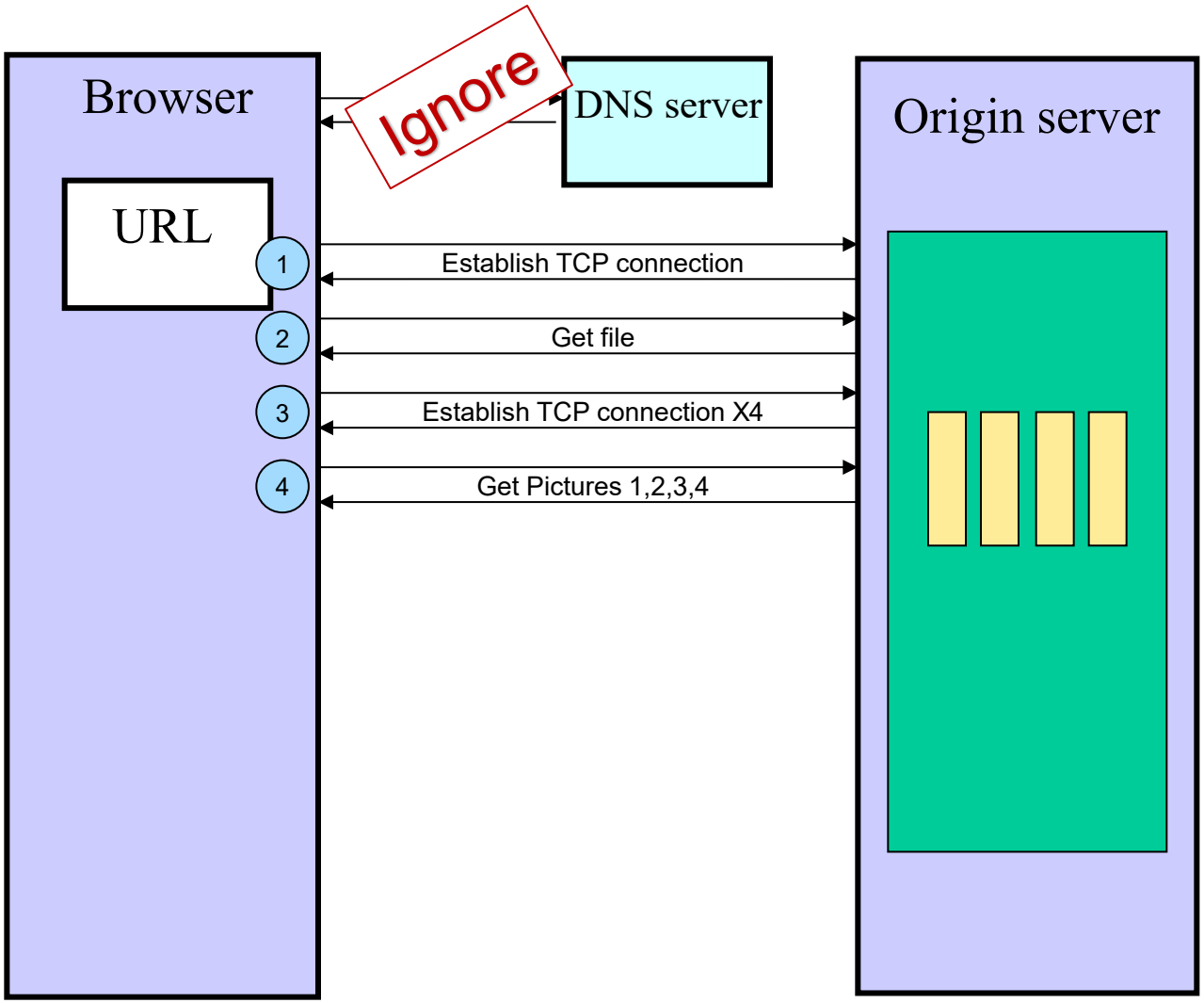
Questions

3. Persistent without pipelining without parallel connections



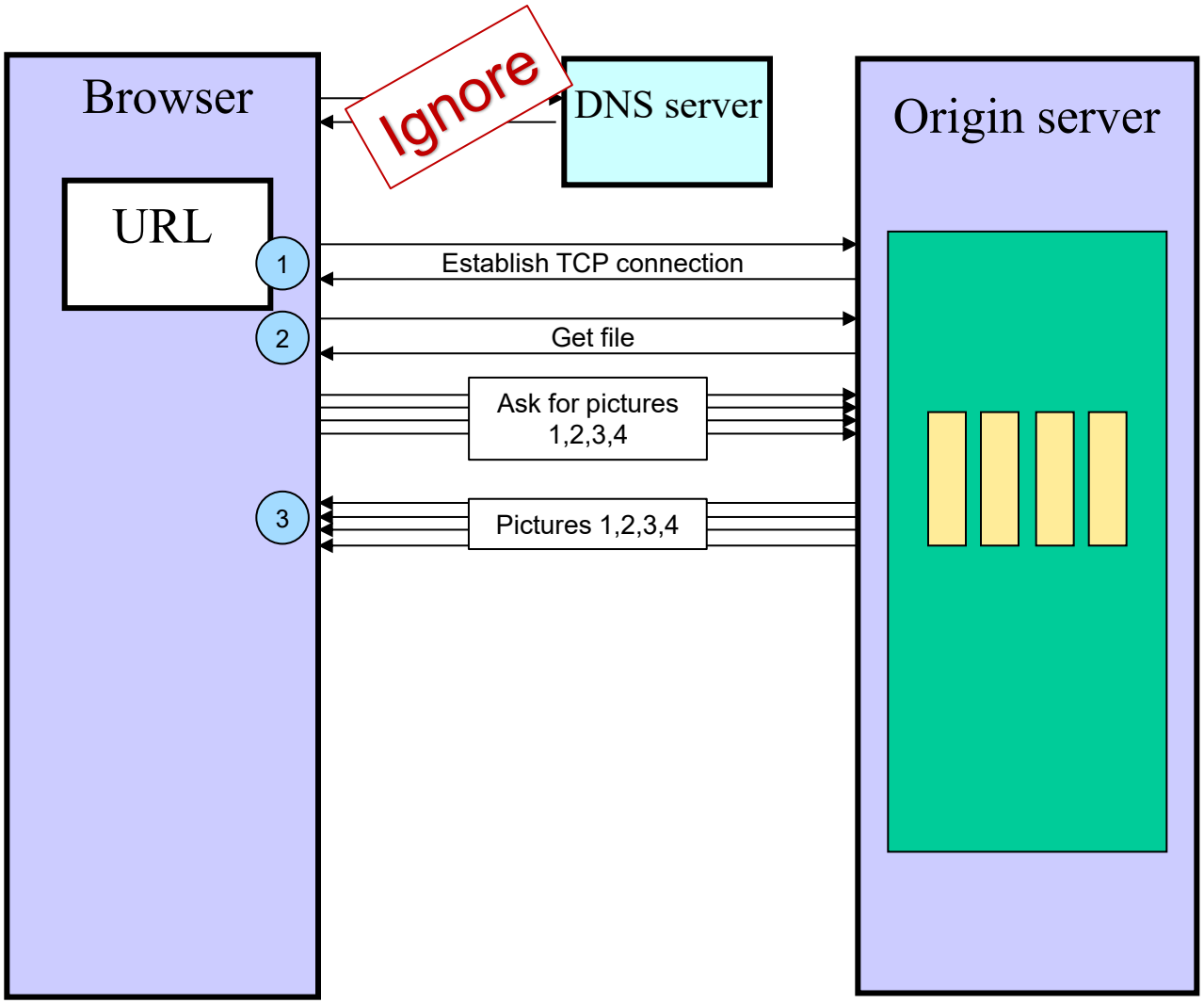
Questions

4. Persistent without pipelining with parallel connections



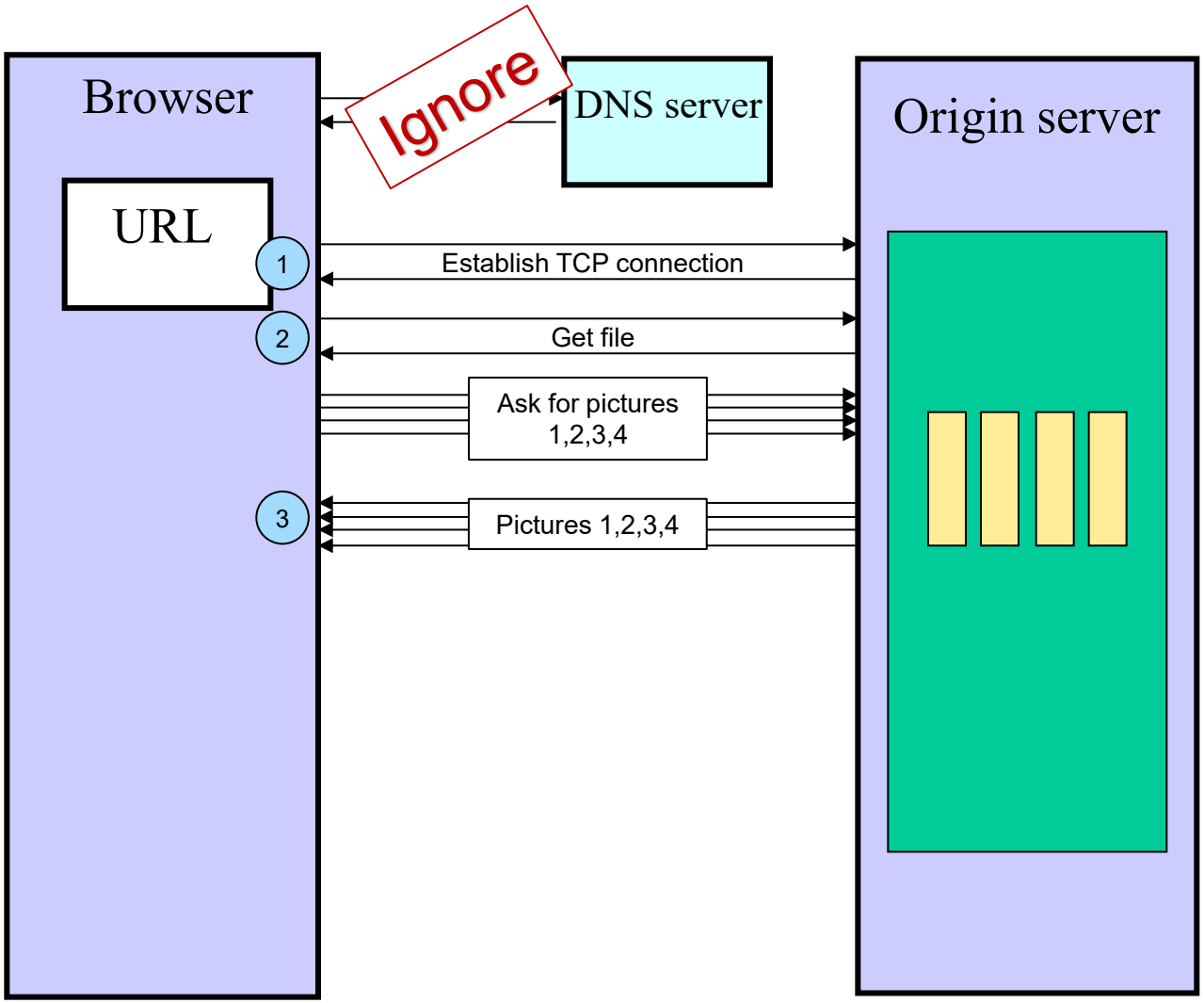
Questions

5. Persistent with pipelining without parallel connections



Questions

6. Persistent with pipelining with parallel connections



Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

אם דף ווב (html) מכיל שתי תמונות והדף ושתי התמונות נמצאים על שלשה שרתים שונים אזי לקשר קבוע **עם** pipelining וללא קשרים מקבילים ולקשר קבוע **ללא** pipelining וללא קשרים מקבילים יש בדיוק אותו זמן תגובה מרגע הבקשה ועד שהדף מגיע.

1. נכון
2. לא נכון

Advance two slides for the English version

Question from an exam
Circle the correct answer

שאלה מבחינה (5 נקודות)
הקף בעיגול את התשובה המתאימה

הנח שלקוח מעונין לקבל דף Web הנמצא בכתובת URL מסוימת. כתובת ה- IP של שרת ה- Web ידועה מראש ללקוח. הדף (html) מכיל **5** תמונות מסוג GIF. רק תמונה אחת נמצאת על השרת בו נמצא הדף (html) המקורי **ו- 4** התמונות האחרות נמצאות על **3** שרתים שונים. כתובות ה- IP של כל השרתים ידועות מראש ללקוח. הנח שהזמן הדרוש ללקוח להתקשר ולקבל תשובה משרת כלשהו הוא RTT אחד. בנוסף, הנח שיש שימוש בקשרים קבועים עם pipelining , ואסור להשתמש בקשרים מקבילים. כמה RTT-ים נחוצים מהרגע בו מכניס הלקוח את ה- URL עד שהדף המלא (כולל התמונות) מוצג ללקוח?

1. ששה
2. שבעה
3. שמונה
4. תשעה
5. כל התשובות (1 עד 4) לא נכונות

Advance two slides for the English version

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

An html webpage refers to two GIF images. The webpage and the two images are located on three different servers. Therefore, a persistent connection **with** pipelining and with no parallel connections, and a persistent connection **without** pipelining and with no parallel connections, have the same response time from the moment of the request until the whole page (with the images) is presented to the user.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (5 נקודות)
הקף בעיגול את התשובה המתאימה

Suppose a client is interested in a web page located in some URL. The IP address of the web server is known in advance by the client. The (html) page consists of **5** GIF images. Only one image is stored on the server where the original (html) page is stored, and the **4** other images are stored on **3** different servers. The IP addresses of all the servers are known in advance to the client. Suppose the time it takes the client to communicate and receive a response from any server is one RTT. In addition, suppose the client is using persistent connections with pipelining, but parallel connections are not allowed. How many RTTs are necessary from the moment the user has typed the URL until the page is fully presented to him (including the images)?

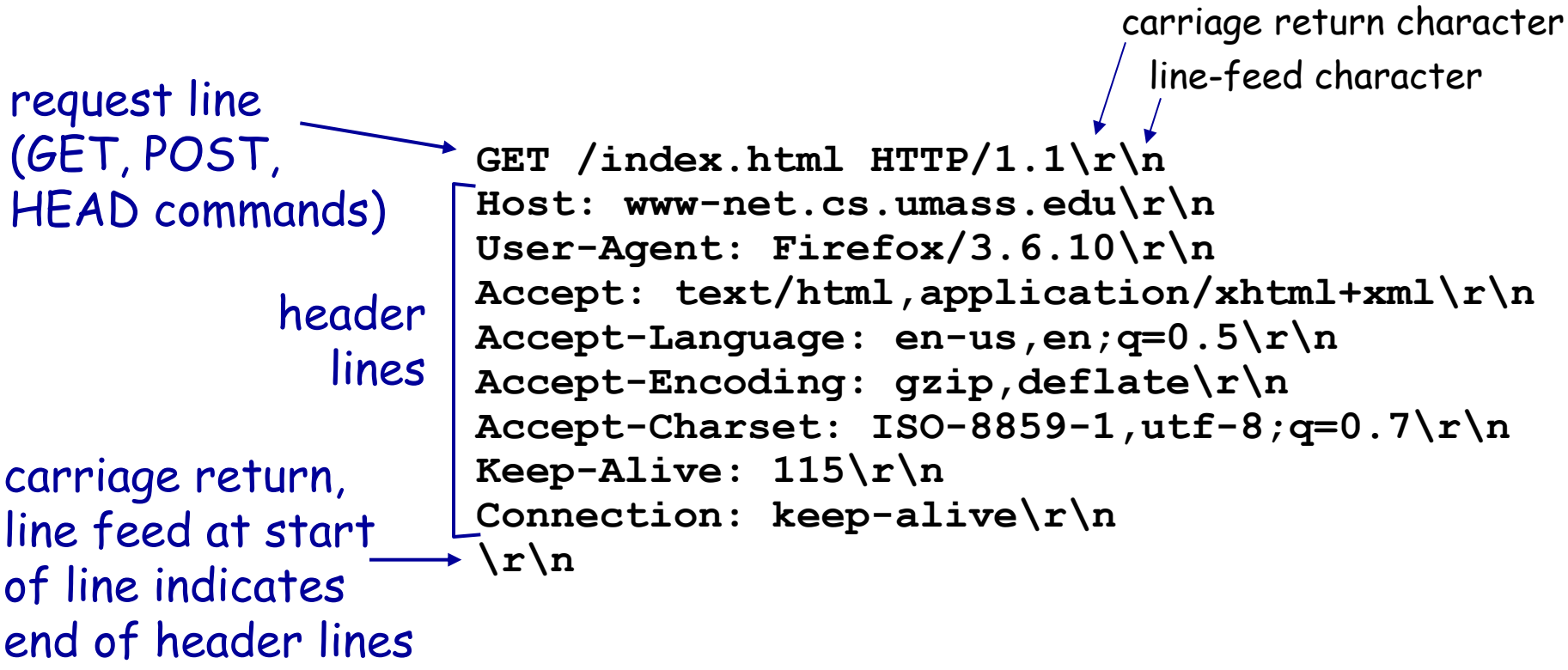
1. Six
2. Seven
3. Eight
4. Nine
5. Answers above (1-4) are incorrect

Today

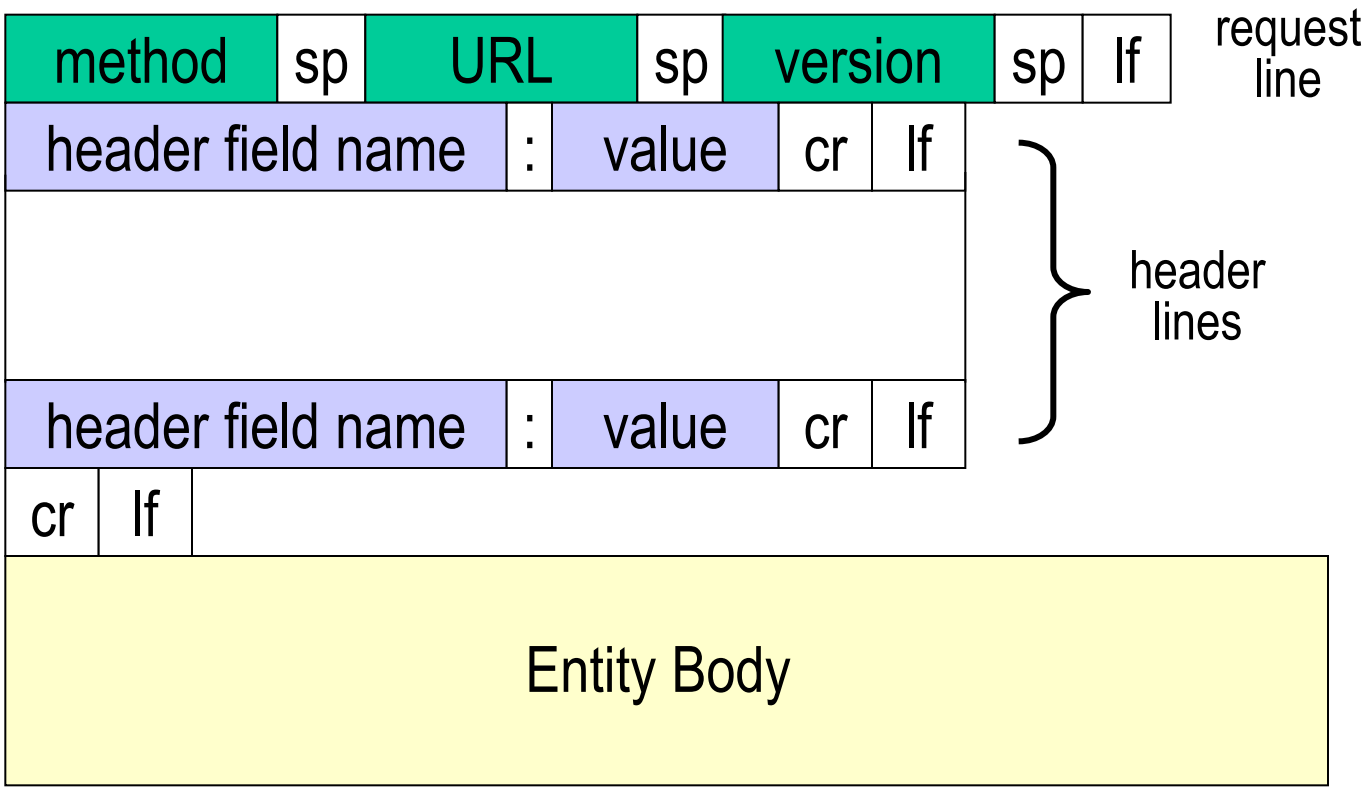
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HTTP request message

- two types of HTTP messages: *request*, *response*
- HTTP request message:
  - ASCII (human-readable format)



http request message: general format

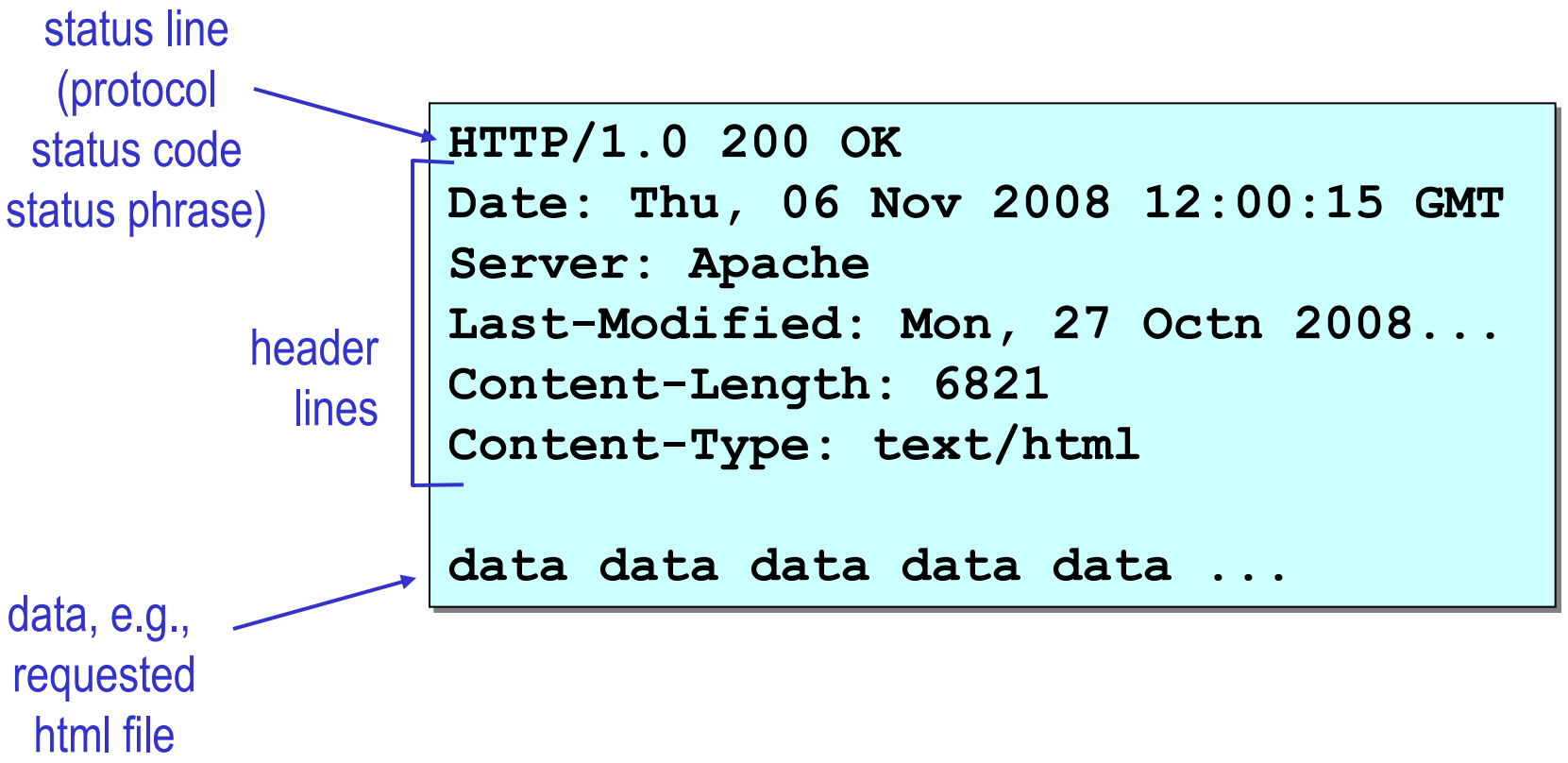



```
GET /somedir/page.html HTTP/1.0
User-agent: Mozilla/4.0
Accept:text/html,image/gif,image/jpeg
Accept-language:fr
```

Request message methods

- GET: used when the browser request an object (most of http request messages use the GET method).
- POST: only when there is an Entity body
  - User fills out a form
  - User give search words to a search engine
- HEAD: used by server developers for debugging (server responds but leaves out the requested object).
- PUT, DELETE
- OPTIONS: example -- max size for put HTTP/1.1
- TRACE: sending back a copy of the message received 1.1
-
```
CONNECT www.example.com:443 HTTP/1.1
(tunneling via proxy … more later)
```

http message format: respone



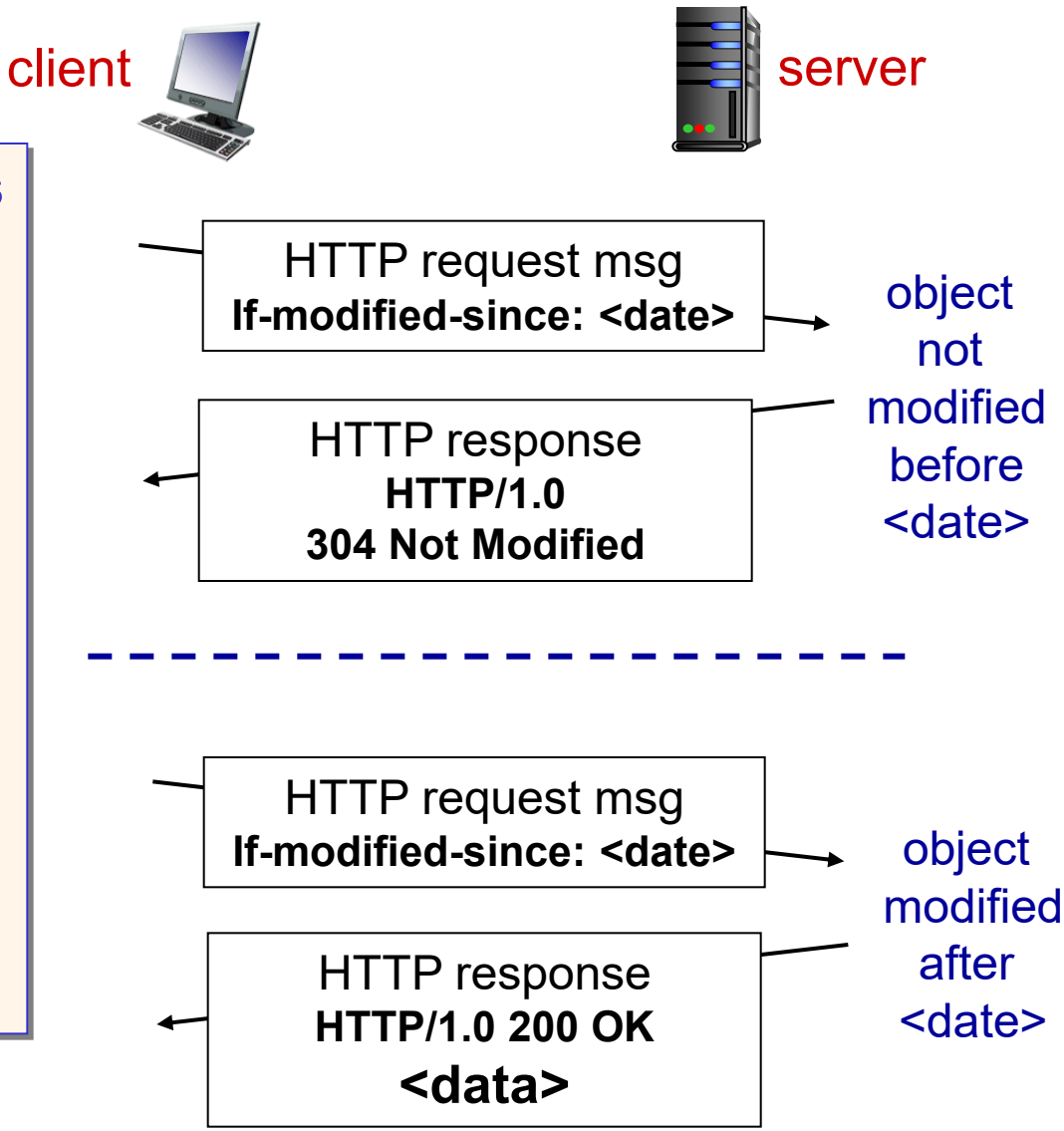
User-server interaction: conditional GET

- Goal: don't send object if client has up-to-date stored (cached) version
- client: specify date of cached copy in http request

```
If-modified-since:
  <date>
```

- server: response contains no object if cached copy up-to-date:

```
HTTP/1.0 304 Not
  Modified
```



Conditional GET

- HTTP Protocol defines a caching mechanism. The request of asking the server for a document from using “IF” header is called Conditional GET Request.
- Using Conditional GET a client can get HTTP response with the content data only if it needs to:

Client:

```
GET /index.html HTTP/1.1
Host: somewhere.com
```

Server:

```
HTTP/1.1 200 OK
Content-Length: 2435
Content-Type: text/html
Date: Tue, 12 Nov 2024 05:25:19 GMT
Cache-Control: max-age=10000
Last-Modified: Sun, 14 Aug 2023 14:20:31 GMT
Etag: "af97bc34"
```

Cache for 10000 seconds

Last modified

Entity tag

Conditional GET

Client:

```
GET / index.html HTTP/1.1
Host: somewhere.com
If-Modified-Since: Sun, 14 Aug 2023 14:20:31 GMT
If-None-Match: "af97bc34"
```

Server:

```
HTTP/1.1 304 Not Modified
Date: Tue, 12 Nov 2024 09:25:19 GMT
Etag: "af97bc34"
Cache-Control: max-age=10000
```

User-server interaction: authentication

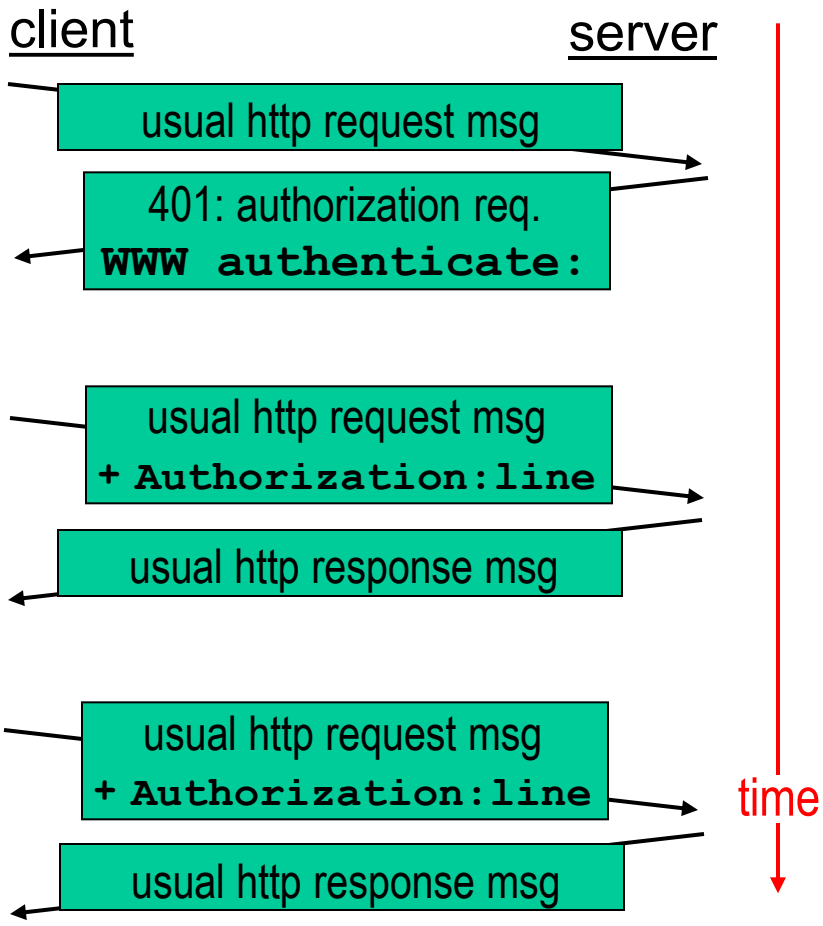
Authentication goal: control access to server documents

- stateless: client must present authorization in each request
- authorization: typically name, password
  - **`authorization:`** header line in request
  - if no authorization presented, server refuses access, sends

    `WWW authenticate:`

    header line in response

Browser caches name & password so that user does not have to repeatedly enter it.



http response status codes

In first line in server to client response message.

A few sample codes:

200 OK

- request succeeded, requested object later in this message

301 Moved Permanently

- requested object moved, new location specified later in this message (Location:)

304 Not Modified

- when using: **If-modified-since**

400 Bad Request

- request message not understood by server

403 Access Denied

- requested document is restricted from current user

404 Not Found

- requested document not found on this server

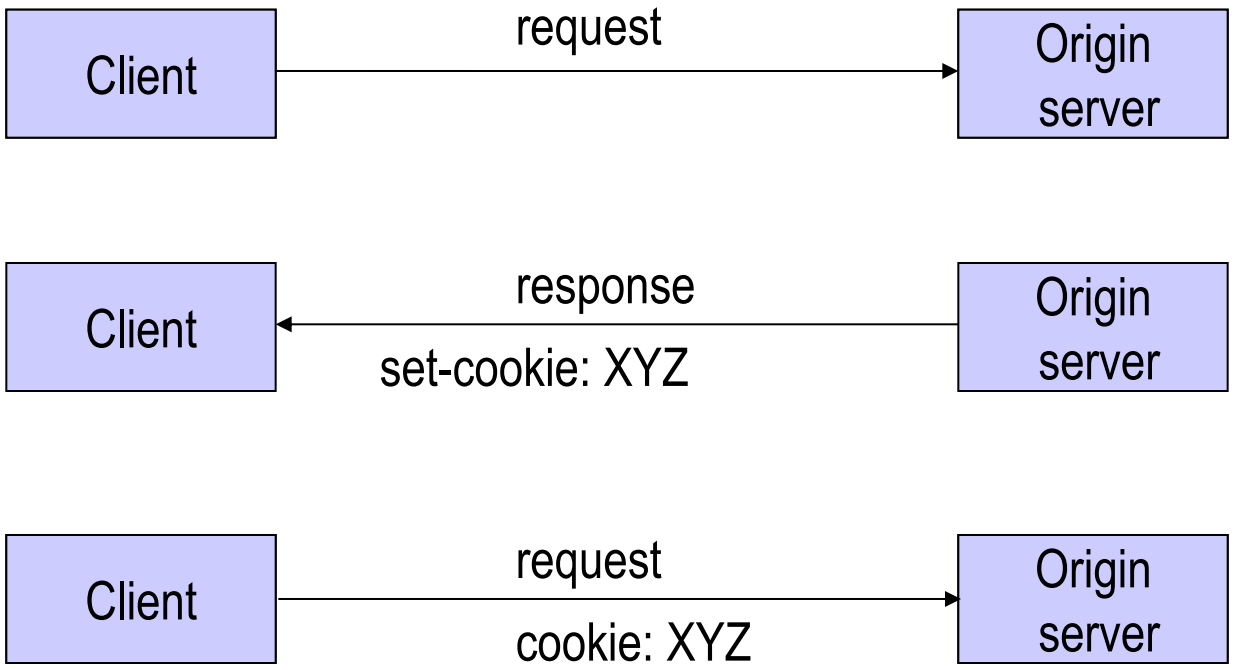
505 HTTP Version Not Supported

A fun short (4:07 min) TED talk about 404 HTTP response.

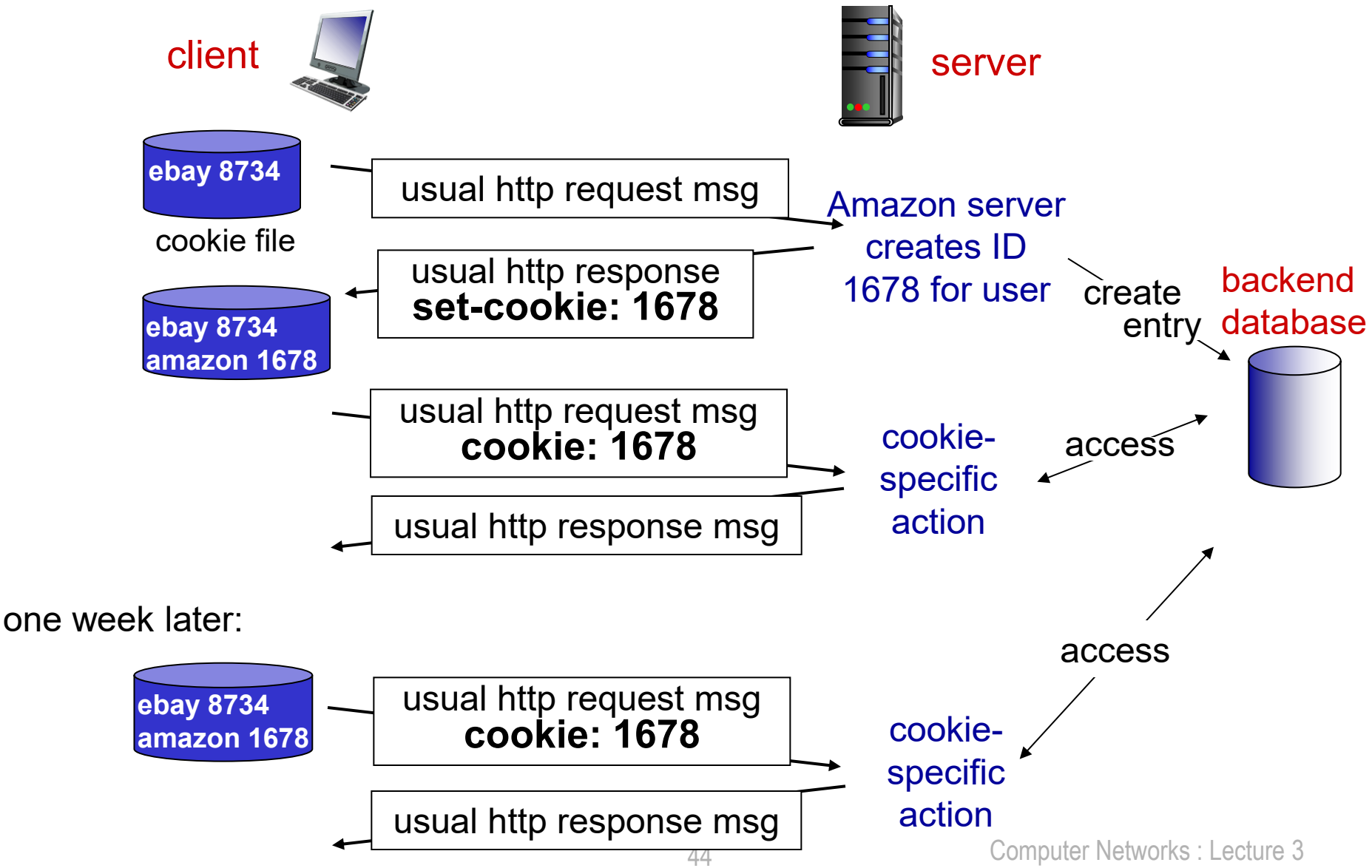
https://www.youtube.com/watch?v=eHrcRqu_Es4

Cookies

Cookies: State information passed between the user agent and the origin server



Cookies


client
server
ebay 8734
cookie file
usual http request msg
Amazon server creates ID 1678 for user
create entry
backend database
usual http response
set-cookie: 1678
ebay 8734
amazon 1678
usual http request msg
cookie: 1678
cookie-specific action
access
usual http response msg
one week later:
access
ebay 8734
amazon 1678
usual http request msg
cookie: 1678
cookie-specific action
usual http response msg

Cookies (continued)

What cookies can bring:

- authorization
- shopping carts (previous choices)
- recommendations
- user session state (Web e-mail)
- remembering user preferences (personal news)

Cookies and privacy:

- cookies permit sites to learn a lot about you
- you may supply name and e-mail to sites
- search engines use redirection & cookies to learn yet more
- advertising companies obtain info across sites

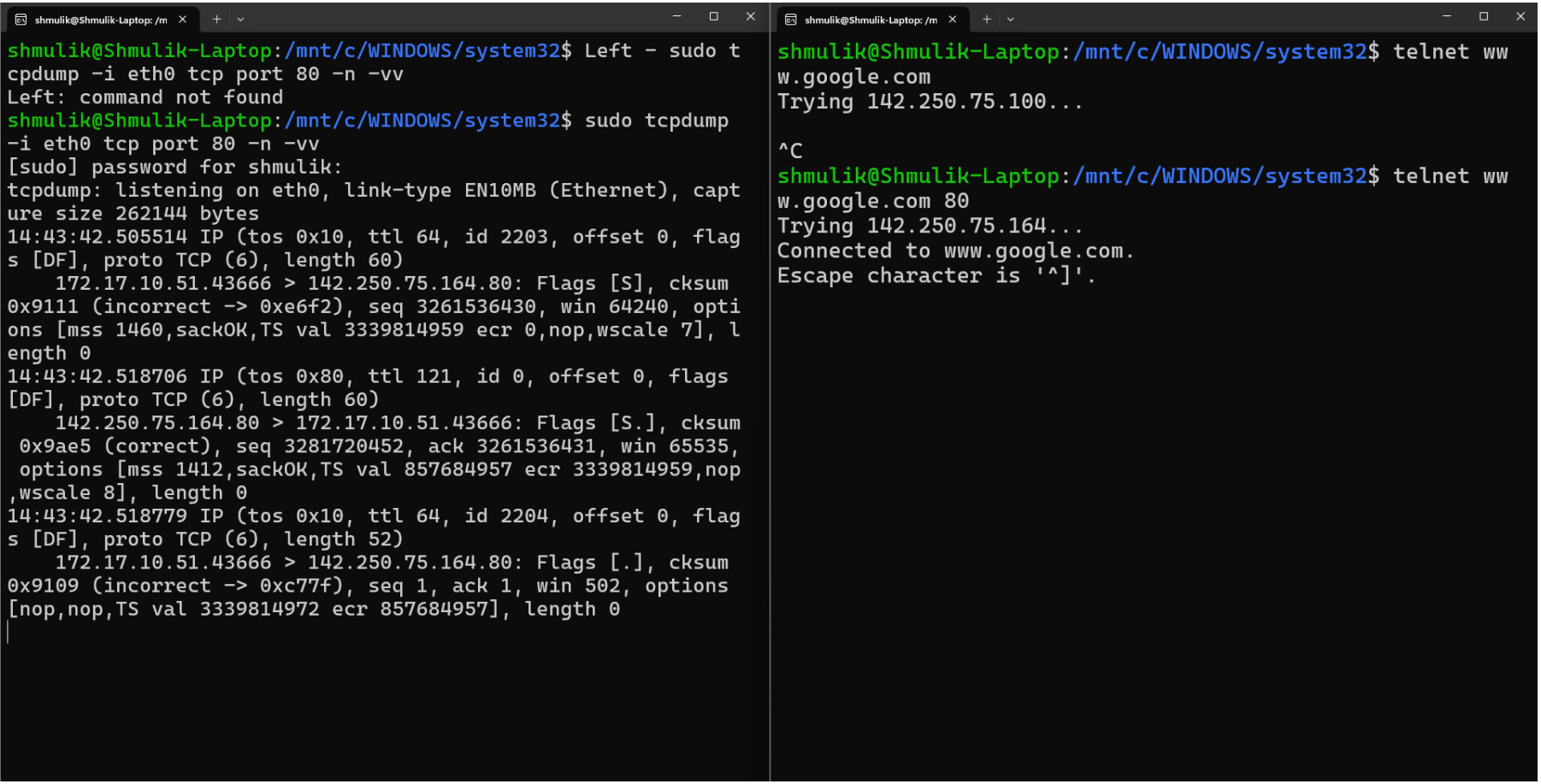
Does cookies make HTTP a stateful protocol?



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HTTP runs on TCP



HTTP runs on TCP

```
14:43:42.505514 IP (tos 0x10, ttl 64, id 2203, offset 0, flags [DF], proto TCP (6), length 60)
    172.17.10.51.43666 > 142.250.75.164.80: Flags [S], cksum 0x9111 (incorrect -> 0xe6f2), seq
    3261536430, win 64240, options [mss 1460,sackOK,TS val 3339814959 ecr 0,nop,wscale 7], length 0
14:43:42.518706 IP (tos 0x80, ttl 121, id 0, offset 0, flags [DF], proto TCP (6), length 60)
    142.250.75.164.80 > 172.17.10.51.43666: Flags [S.], cksum 0x9ae5 (correct), seq 3281720452, ack
    3261536431, win 65535, options [mss 1412,sackOK,TS val 857684957 ecr 3339814959,nop,wscale 8], length
    0
14:43:42.518779 IP (tos 0x10, ttl 64, id 2204, offset 0, flags [DF], proto TCP (6), length 52)
    172.17.10.51.43666 > 142.250.75.164.80: Flags [.], cksum 0x9109 (incorrect -> 0xc77f), seq 1, ack 1,
    win 502, options [nop,nop,TS val 3339814972 ecr 857684957], length 0
```

HEAD & Response

```
shmulik@Shmulik-Laptop:/mnt/c/WINDOWS/system32$ telnet www.google.com 80
Trying 142.250.75.100...
Connected to www.google.com.
Escape character is '^]'.
HEAD / HTTP/1.1
Host: www.google.com
user-agent: demo-client/1.0
accept: */*
Accept-Language: en-US,en;q=0.9
Connection: close

HTTP/1.1 200 OK
Content-Type: text/html; charset=ISO-8859-1
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-MdhiqsI0qH_G3m2BADlOEA' 'stri
ct-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/o
ther-hp
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Sat, 08 Nov 2025 12:49:14 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Sat, 08 Nov 2025 12:49:14 GMT
Cache-Control: private
Set-Cookie: AEC=AaJma5vQd9eRlnDGZ2FJnBSaZQBiNgN1hOoss4hfb-Ch0b1lM3Rh9YJDDw; expires=Thu, 07-May-2026 12:49:14 GMT; pat
h=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=oxLr4pLc5XMnHBDLhoWt4L_6p6--bT6CPk7b3vXAwywwIc90m3uT7a8oYS09HIQn4UuLeCO3wzFaRdLEyDxvIeSJ1Pw12bRMTG
_Se6suCLtJL6TVkywFvmhfu7FXT0QGw7rwU1vcU-CC6rusAMGYPc2n0PV_OljKDXT4jCcuLSshA_p2893S9sRjTtzQYb48bAhVZ6hVpxaJVQ; expires=
Sun, 10-May-2026 12:49:14 GMT; path=/; domain=.google.com; HttpOnly
Transfer-Encoding: chunked
Connection: close

Connection closed by foreign host.
shmulik@Shmulik-Laptop:/mnt/c/WINDOWS/system32$
```

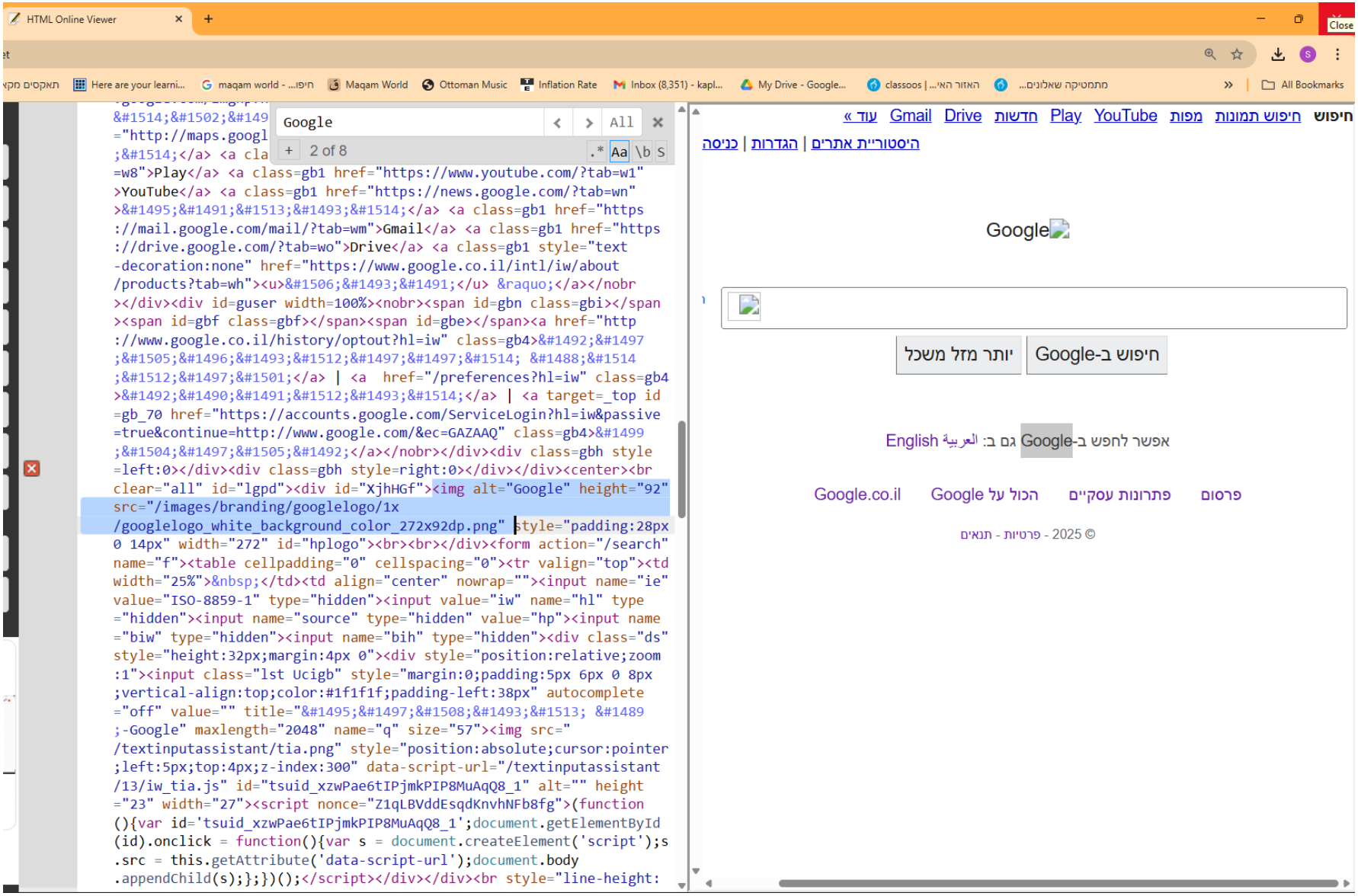
GET “/” from www.google.com

```
shmulik@Shmulik-Laptop:/mnt/c/WINDOWS/system32$ telnet www.google.com 80
Trying 142.250.75.100...
Connected to www.google.com.
Escape character is '^]'.
GET / HTTP/1.1
Host: www.google.com
user-agent: demo-client/1.0
accept: */*
Accept-Language: en-US,en;q=0.9
Connection: keep-alive

HTTP/1.1 200 OK
Date: Sat, 08 Nov 2025 12:51:19 GMT
Expires: -1
Cache-Control: private, max-age=0
Content-Type: text/html; charset=ISO-8859-1
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-Z1qLBVddEsqdKnvhNFb8fg' 'strict-dyna
mic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other-hp
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Set-Cookie: __Secure-STRP=ADq1D7pUyYVT-DyvV_T1sSesxmmZT2lL-HrP_RlvPFeKYNyOJmSpg7kPQNq_RAuun2DRPeQLcFPrK8rxPhjXIlpmIjE8ZTA82vk
K; path=/; domain=.google.com; Secure; SameSite=strict
Set-Cookie: AEC=AaJma5vkTSj4I433ulsGWRpHjY0OPTKwyGIQVU_vKn8qdeEJCwJt242t5a4; expires=Thu, 07-May-2026 12:51:19 GMT; path=/; d
omain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=YrZLq-u1OPljBWr78vNzUe-pXVT24MKcngfj7Se1ZrJd6YWDU2n89PYNN7OBiBpju5NtjA7FF99ih21-Nc0nv386y306YGDhBW8yjmLVO
imjZetYGhvGuNIhna5mPMxXN6TrsRcFnXcA-rR7tcWQ6g4dVZuACWnL2Nh_yvy88IoeSFFKxXBLraGYP1xQoC7H-vEWfp7pzQx9uztfmL0; expires=Sun, 10-M
ay-2026 12:51:19 GMT; path=/; domain=.google.com; HttpOnly
Accept-Ranges: none
Vary: Accept-Encoding
Transfer-Encoding: chunked

75f
<!doctype html><html dir="rtl" itemscope="" itemtype="http://schema.org/WebPage" lang="iw"><head><meta content="text/html; ch
```

www.google.com – response (html)



Asking for an image

/images/branding/googlelogo/1x/googlelogo_white_background_color_272x92dp.png

```
shmulik@Shmulik-Laptop:/mnt/c/WINDOWS/system32$ telnet www.google.com 80
Trying 142.250.75.100...
Connected to www.google.com.
Escape character is '^]'.
GET /images/branding/googlelogo/1x/googlelogo_white_background_color_272x92dp.png HTTP/1.1
Host: www.google.com
user-agent: demo-client/1.0
accept: */*
Accept-Language: en-US,en;q=0.9
Connection: keep-alive

HTTP/1.1 200 OK
Accept-Ranges: bytes
Content-Type: image/png
Cross-Origin-Resource-Policy: cross-origin
Cross-Origin-Opener-Policy-Report-Only: same-origin; report-to="stat
Report-To: {"group":"static-on-bigtable","max_age":2592000,"endpoint
ic-on-bigtable"}]}
Content-Length: 5482
Date: Sat, 08 Nov 2025 12:56:13 GMT
Expires: Sat, 08 Nov 2025 12:56:13 GMT
Cache-Control: private, max-age=31536000
Last-Modified: Tue, 22 Oct 2019 18:30:00 GMT
X-Content-Type-Options: nosniff
Server: sffe
X-XSS-Protection: 0

PNG
IHDR)}1IDATx]Ȳg0?f
23333fwL2`{<üzK-Kr:6-]uuv
                                                    ,G W yc y W CW #W 6x v G d`P^ 05g p s{ ń {w K
                                                                                                  a* j z y 8{
.a{6a T
:[A r{81r o, :zQAY \g V, SC c]?
```

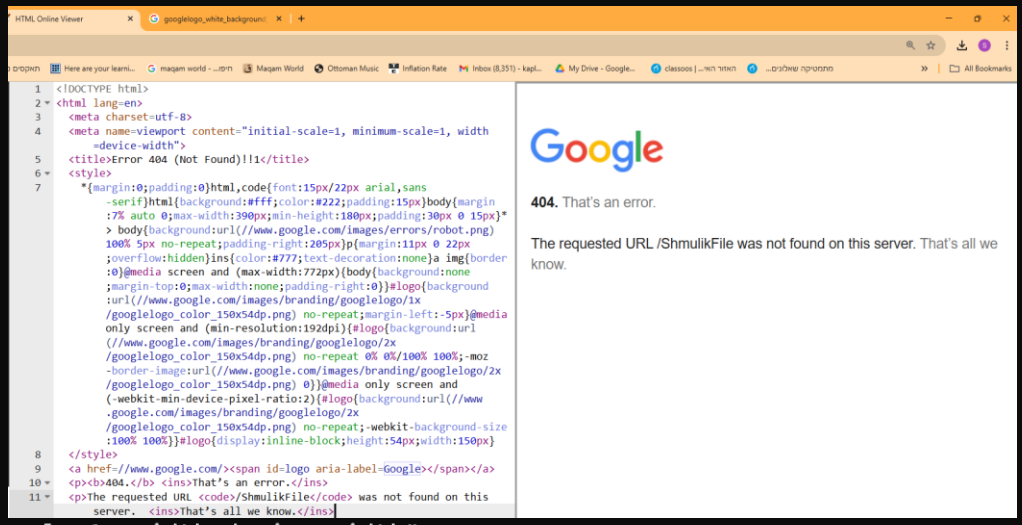


404 – Not Found

```
shmulik@Shmulik-Laptop:/mnt/c/WINDOWS/system32$ telnet www.google.com 80
Trying 142.250.75.100...
Connected to www.google.com.
Escape character is '^]'.
GET /ShmulikFile HTTP/1.1
Host: www.google.com
user-agent: demo-client/1.0
accept: */*
Accept-Language: en-US,en;q=0.9
Connection: keep-alive

HTTP/1.1 404 Not Found
Content-Type: text/html; charset=UTF-8
Referrer-Policy: no-referrer
Content-Length: 1572
Date: Sat, 08 Nov 2025 12:59:12 GMT

<!DOCTYPE html>
<html lang=en>
  <meta charset=utf-8>
  <meta name=viewport content="initial-scale=1, minimum-scale=1, width=device-width">
  <title>Error 404 (Not Found)!!1</title>
  <style>
    *{margin:0;padding:0}html,code{font:15px/22px arial,sans-serif}html{background:#fff;color:#222;padding:15px}body{margin:7
% auto 0;max-width:390px;min-height:180px;padding:30px 0 15px}* > body{background:url(//www.google.com/images/errors/robot.pn
g) 100% 5px no-repeat;padding-right:205px}p{margin:11px 0 22px;overflow:hidden}ins{color:#777;text-decoration:none}a img{bord
er:0}@media screen and (max-width:772px){body{background:none;margin-top:0;max-width:none;padding-right:0}}#logo{background:u
rl(//www.google.com/images/branding/googlelogo/1x/googlelogo_color_150x54dp.png) no-repeat;margin-left:-5px}@media only scree
n and (min-resolution:192dpi){#logo{background:url(//www.google.com/images/branding/googlelogo/2x/googlelogo_color_150x54dp.p
ng) no-repeat 0% 0%/100% 100%;-moz-border-image:url(//www.google.com/images/branding/googlelogo/2x/googlelogo_color_150x54dp.
png) 0}}@media only screen and (-webkit-min-device-pixel-ratio:2){#logo{background:url(//www.google.com/images/branding/googl
elogo/2x/googlelogo_color_150x54dp.png) no-repeat;-webkit-background-size:100% 100%}}#logo{display:inline-block;height:54px;w
idth:150px}
  </style>
  <a href=//www.google.com/><span id=logo aria-label=Google></span></a>
```



If-Not-Modified & Response 304

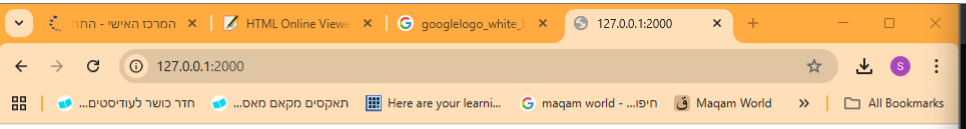
shmulik@Shmulik-Laptop: /m

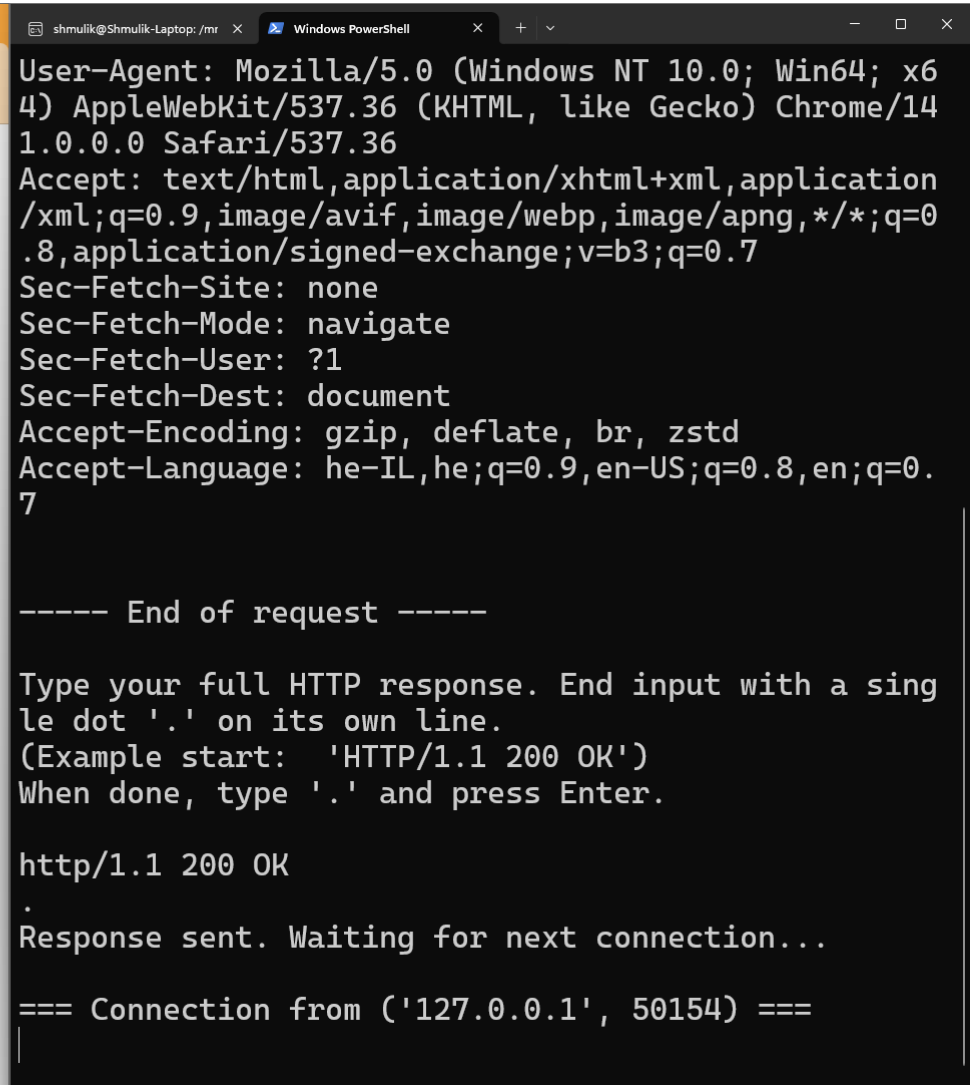
```
png) 0}}@media only screen and (-webkit-min-device-pixel-ratio:2){#logo{background:url(//www.google.com/images/branding/googlelogo/2x/googlelogo_color_150x54dp.png) no-repeat;-webkit-background-size:100% 100%}}#logo{display:inline-block;height:54px;width:150px}
  </style>
  <a href=//www.google.com/><span id=logo aria-label=Google></span></a>
  <p><b>404.</b> <ins>That’s an error.</ins>
  <p>The requested URL <code>/ShmulikFile</code> was not found on this server.  <ins>That’s all we know.</ins>
```

```
GET /images/branding/googlelogo/1x/googlelogo_white_background_color_272x92dp.png HTTP/1.1
Host: www.google.com
user-agent: demo-client/1.0
accept: */*
Accept-Language: en-US,en;q=0.9
Connection: keep-alive
If-Modified-Since: Sat, 08 Nov 2025 11:22:40 GMT

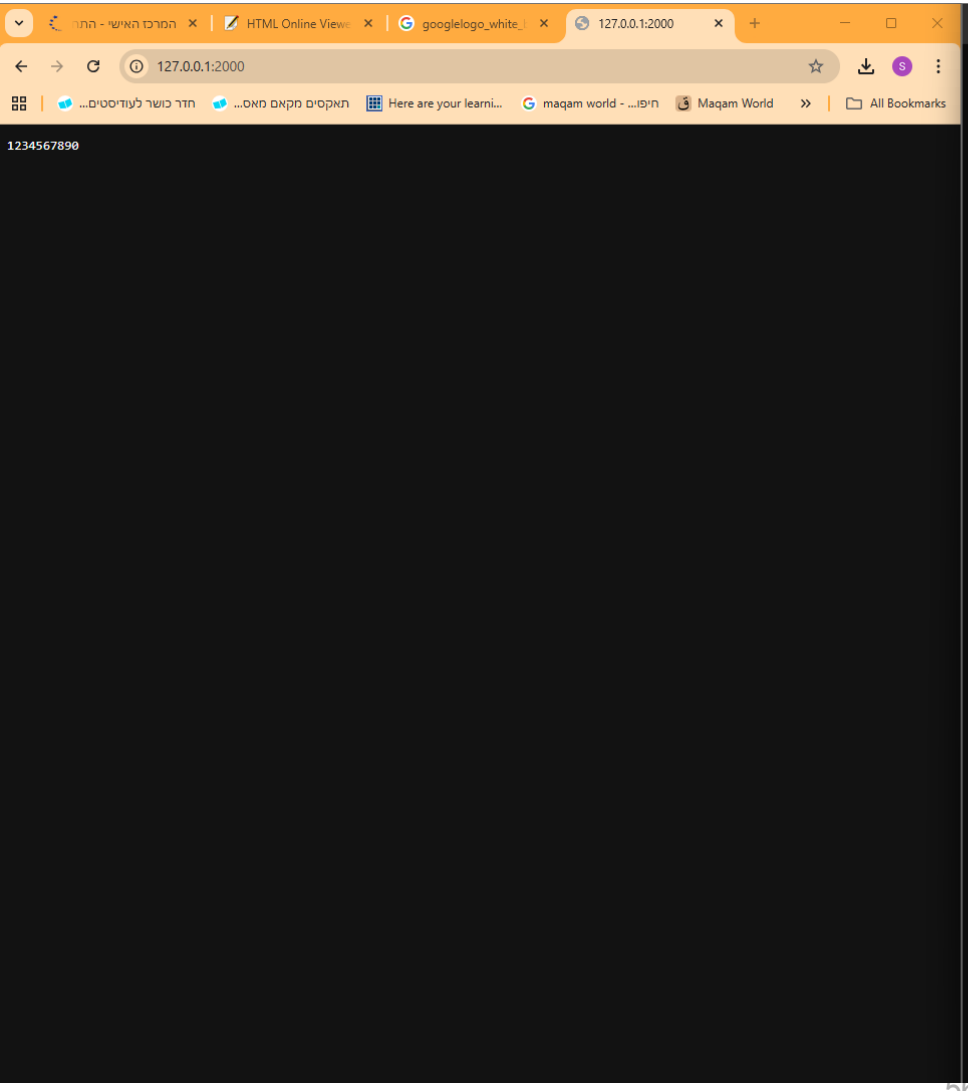
HTTP/1.1 304 Not Modified
Cross-Origin-Resource-Policy: cross-origin
Date: Sat, 08 Nov 2025 13:01:04 GMT
Expires: Sat, 08 Nov 2025 13:01:04 GMT
Cache-Control: private, max-age=31536000
Last-Modified: Tue, 22 Oct 2019 18:30:00 GMT
X-Content-Type-Options: nosniff
Server: sffe
X-XSS-Protection: 0
```

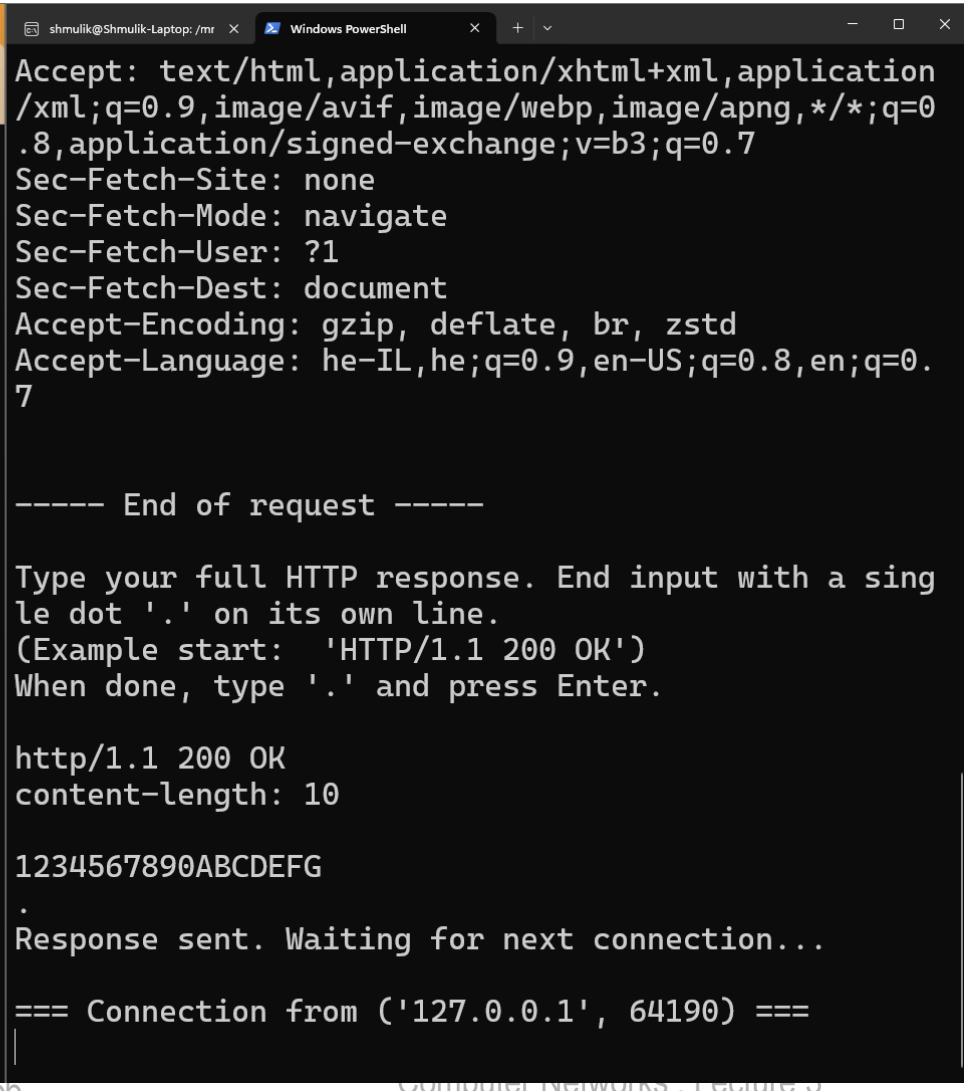
My server (port 2000) – empty 200 OK



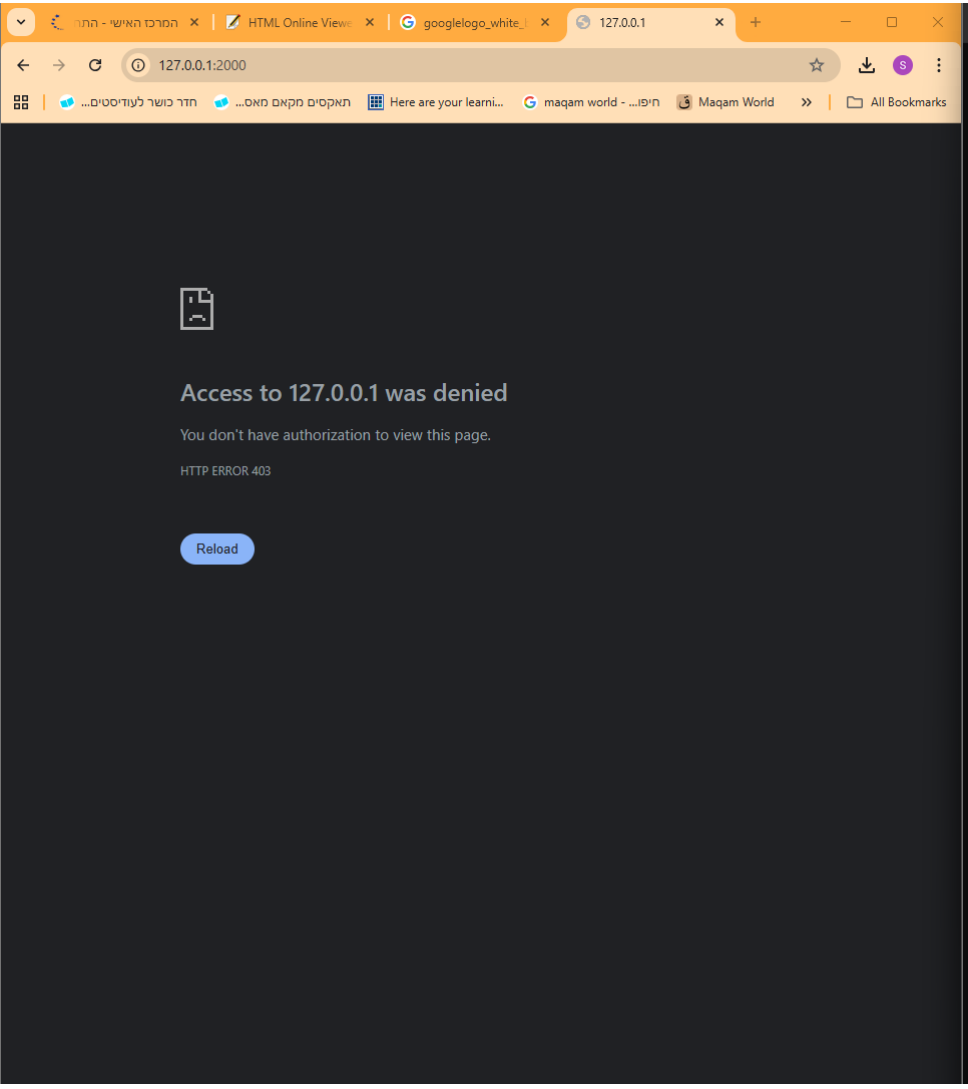


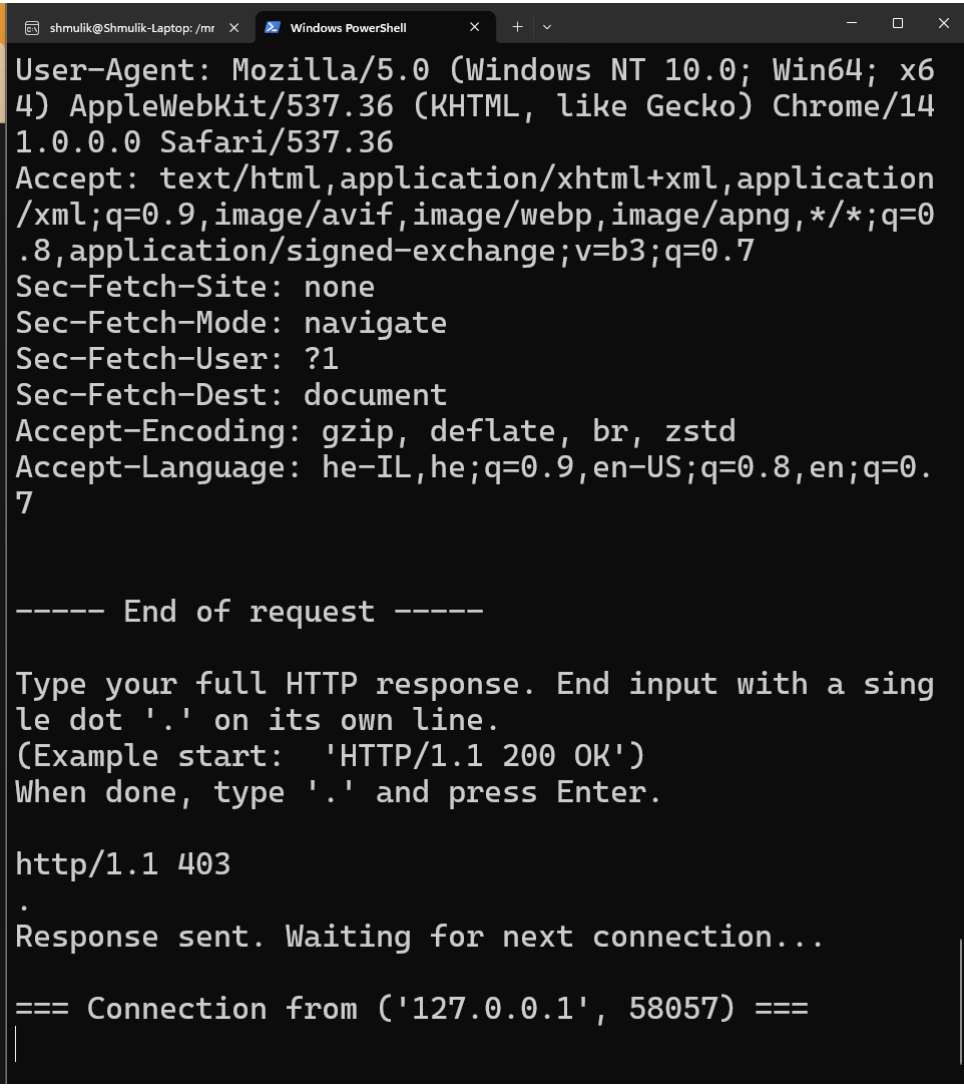
My server (port 2000) – 200 OK – with 10 characters text





My server (port 2000) – 403 ACCESS DENIED





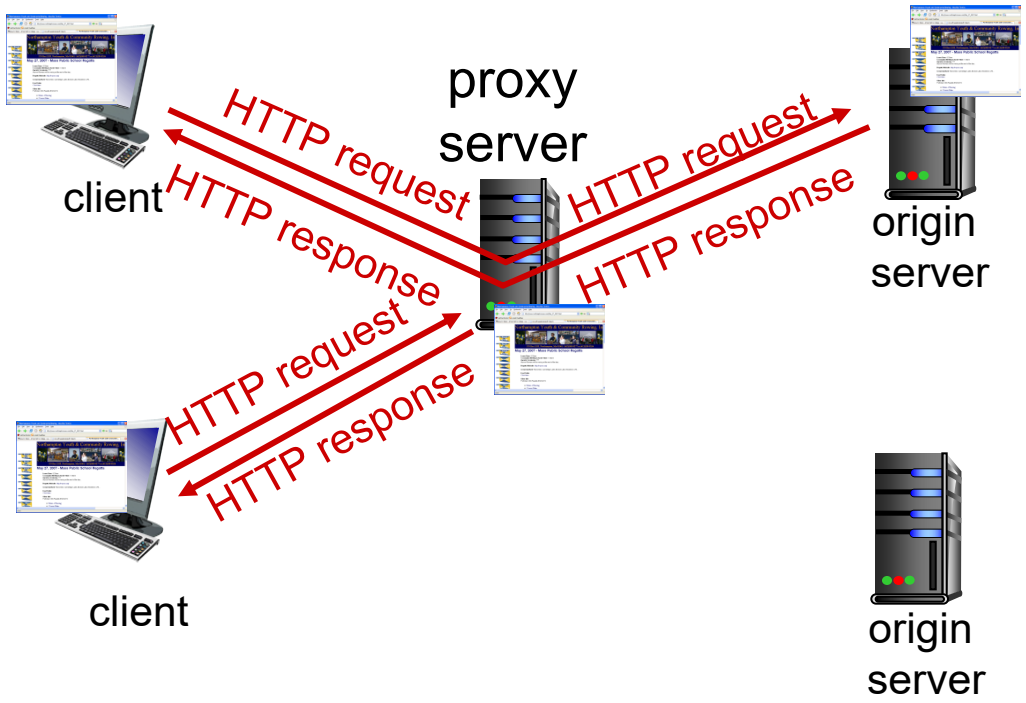
Today

- ✓ Review
- ✓ URL
- ✓ Web and HTTP
- ✓ RTT
- ✓ Questions
- ✓ HTTP definition
  - Request & Response
  - Conditionals
  - Authentication
  - Cookies
- ✓ Demonstration
- Proxy
- Security
- HTTP 2 & HTTP 3

Web caches (proxy server)

Goal: satisfy client request without involving origin server

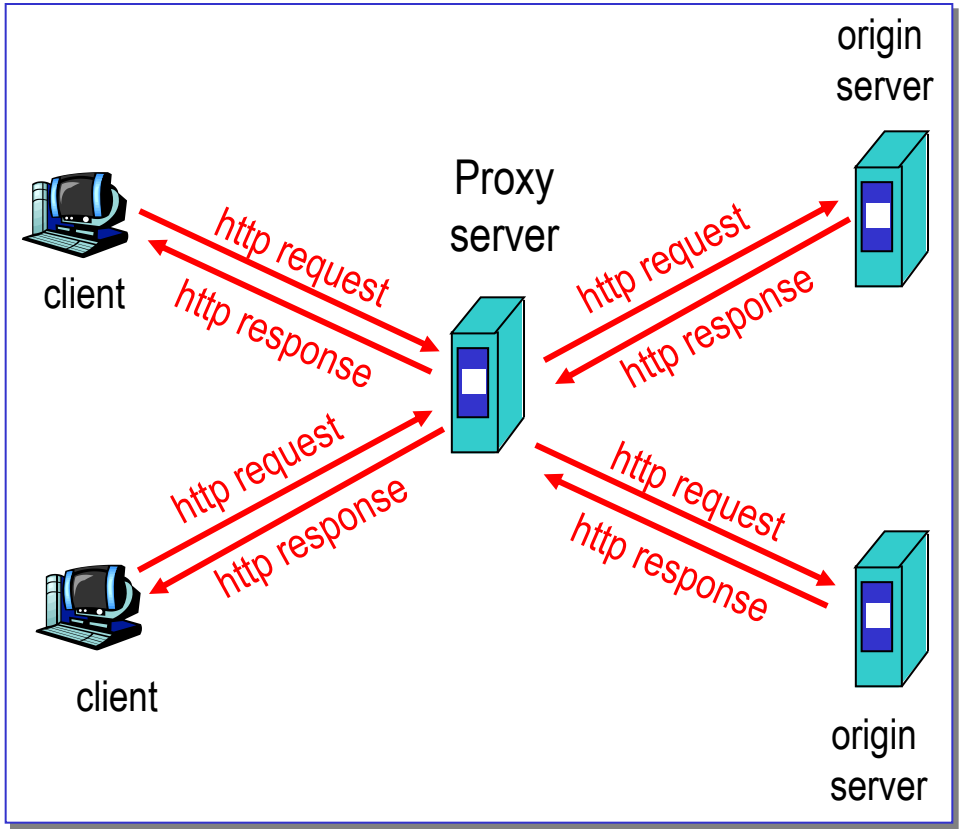
- user sets browser: Web accesses via cache
- browser sends all HTTP requests to cache
  - object in cache: cache returns object
  - else cache requests object from origin server, then returns object to client



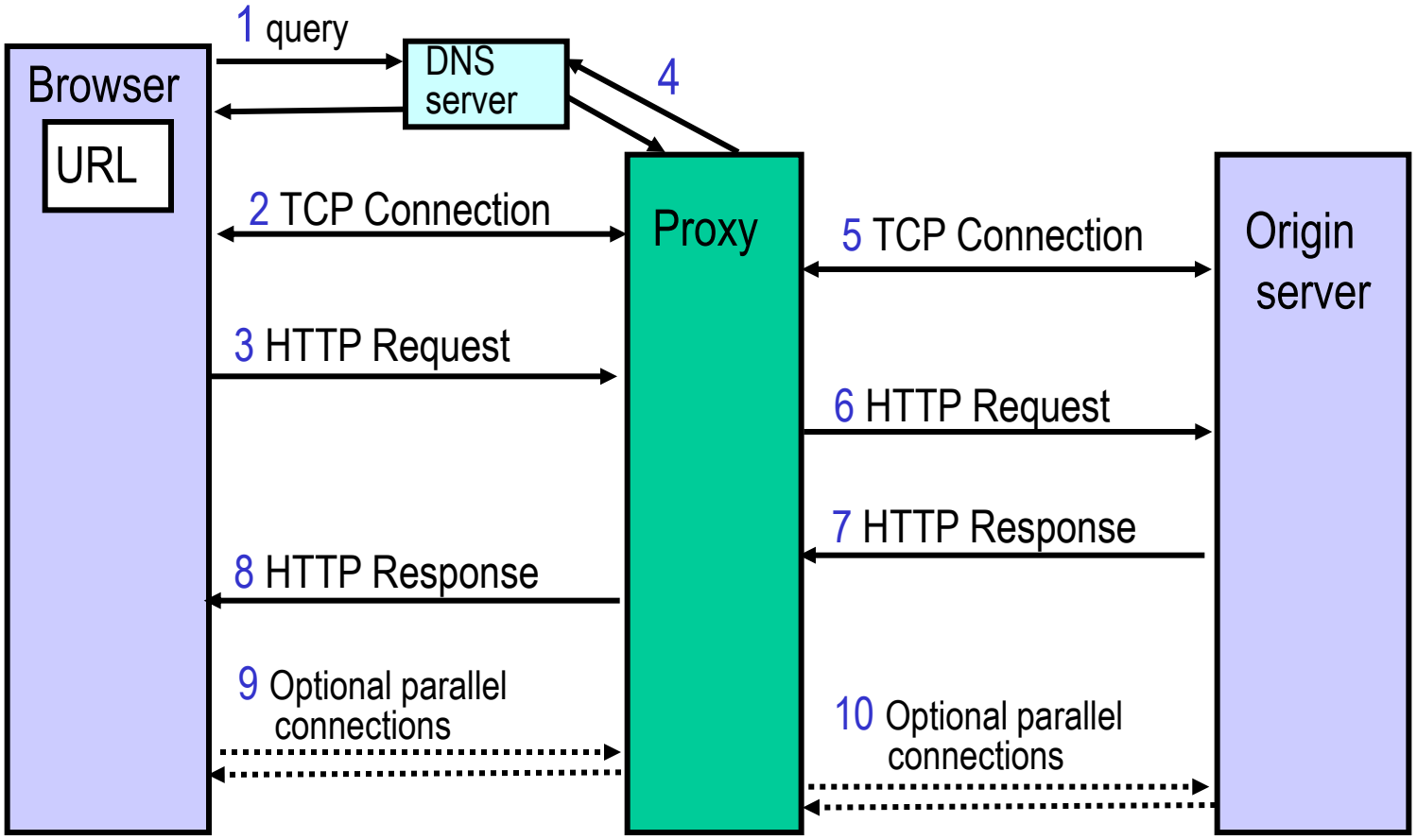
Proxy server

Proxy: A program in the path between the user agent and an origin server that functions as server to the user agent and as client to the origin server

- Caching responses
- Anonymizing clients
- Transforming requests and resp. (low-bandwidth,compression…)
- Gateway to non-HTTP systems
- Filtering requests and responses (URL filtering, searching, remove headers)
- Firewall



Steps in a browser process with proxy in the path




```
Can use also: CONNECT www.example.com:443 HTTP/1.1
(Only the initial connection request is HTTP)
```

Today

- ✓ Review
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  - Conditionals
  - Authentication
  - Cookies
- ✓ Demonstration
- ✓ Proxy
- Security
- HTTP 2 & HTTP 3

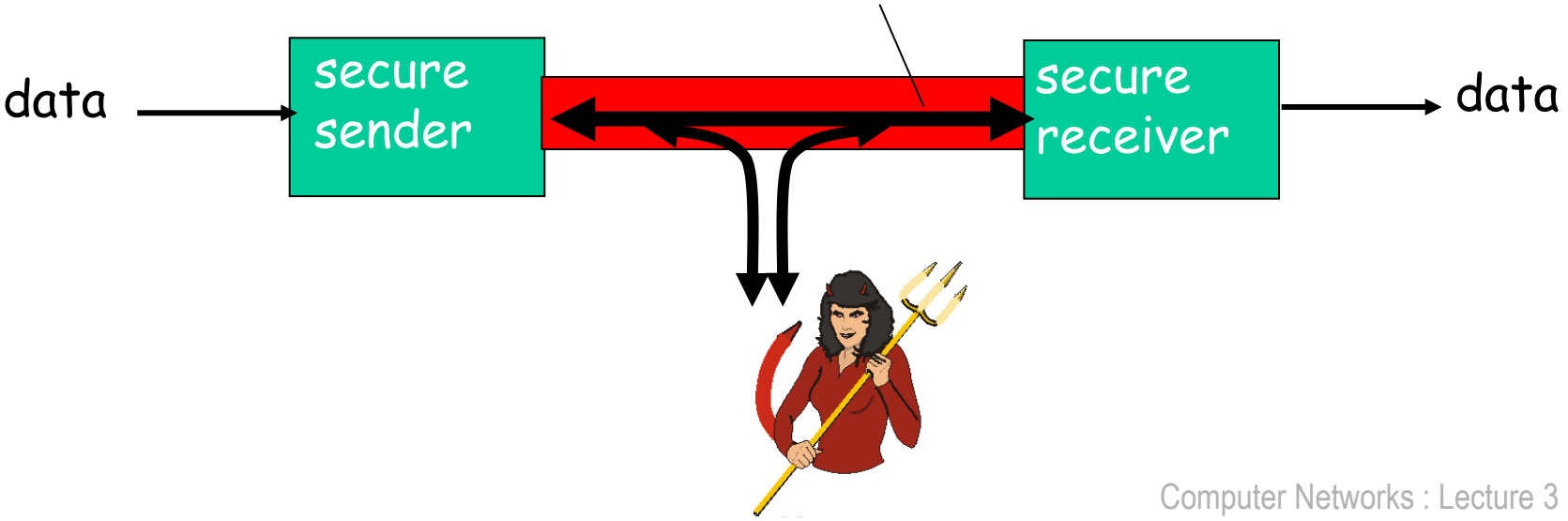
What is network security?

Confidentiality: (סודיות, חשאיות) only sender, intended receiver should "understand" message contents

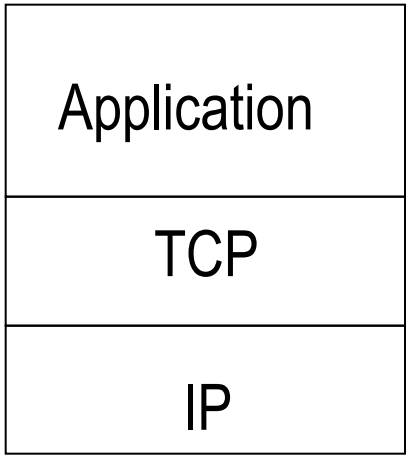
- sender encrypts message
- receiver decrypts message

Authentication: (אימות) sender, receiver want to confirm identity of each other

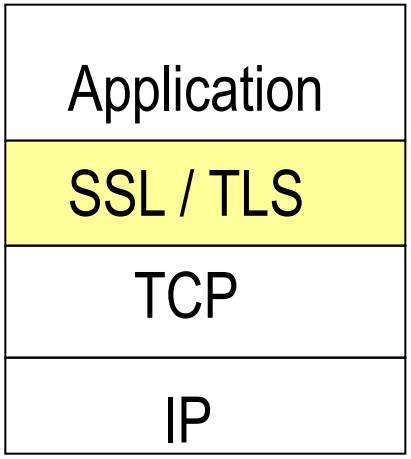
Message integrity: (שלמות) sender, receiver want to ensure message not altered (in transit, or afterwards) without detection.



SSL: Secure Sockets Layer
TLS: Transport Layer Security



Normal Application



Application with SSL

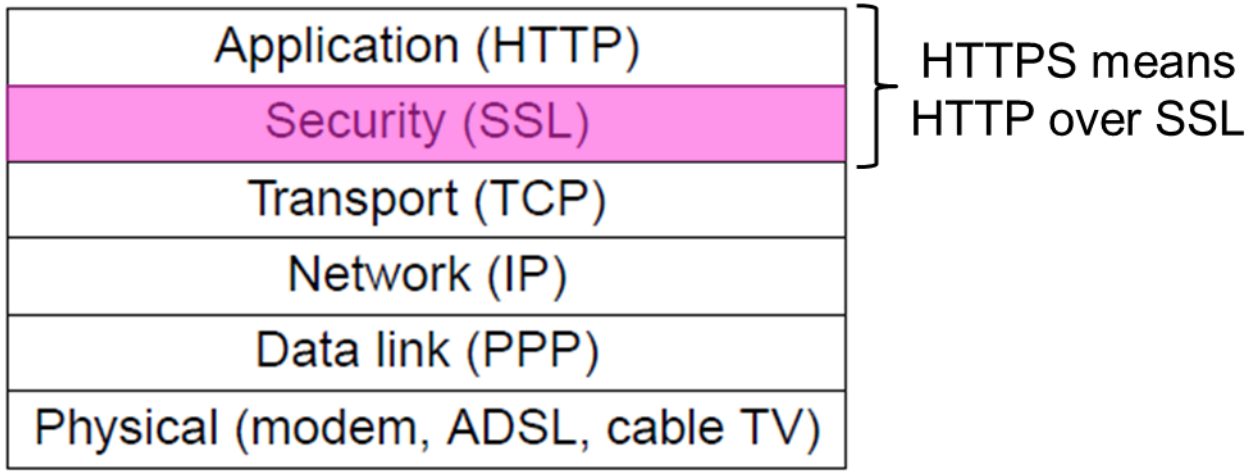
- SSL/TLS provides application programming interface (API) to applications
- C and Java SSL/TLS libraries/classes readily available
- HTTP which uses SSL/TLS is called HTTPs

SSL/TLS

- Widely deployed security protocol
  - supported by almost all browsers, web servers
- Available to all TCP applications

- Provides
  - *confidentiality*
  - *authentication*
  - *integrity*

SSL runs on top of TCP and below the application →



SSL in the protocol stack

Today

- ✓ Review
- ✓ URL
- ✓ Web and HTTP
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- ✓ Questions
- ✓ HTTP definition
  - Request & Response
  - Conditionals
  - Authentication
  - Cookies
- ✓ Demonstration
- ✓ Proxy
- ✓ Security
- HTTP 2 & HTTP 3

HTTP 1.1 vs 2 vs 3

Feature	HTTP/1.1	HTTP/2	HTTP/3
Transport Protocol	TCP	TCP	QUIC (over UDP)
Connection Setup	1 RTT for TCP + 1 RTT for TLS	1 RTT for TCP + 1 RTT for TLS	1 RTT (or 0-RTT) QUIC handshake
Multiplexing	Not really	Yes	Yes
Header Compression	None	HPACK	QPACK
Data Framing	Text-based	Binary frames	Binary frames
Encryption	Optional	Usually encrypted	Always TLS 1.3 built in
Prioritization	Limited	Stream prioritization	Stream prioritization
Loss Recovery	Handled by TCP	Handled by TCP	Handled per-stream by QUIC (no global stalls)
Performance	Baseline, sequential requests	Faster, fewer connections	Fastest, lower latency and mobile-friendly
Browser Support	Universal	Universal	Modern browsers (Chrome, Edge, Firefox, Safari)